



AllLife Bank - Case Study

Project Report

Project Details:

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1. Project Background

AllLife Bank wants to focus on its credit card customer base in the next financial year. They have been advised by their marketing research team, that the penetration in the market can be improved. Based on this input, the Marketing team proposes to run personalized campaigns to target new customers as well as upsell to existing customers.

Another insight from the market research was that the customers perceive the support services of the bank poorly. This suggests that customers have a negative perception of the support services provided by the business or organization. This could mean that customers find the support services to be inadequate, unhelpful, slow, or lacking in quality.

Based on this, the Operations team wants to upgrade the service delivery model, to ensure that customer queries are resolved faster.

2. Objective

Our main objective is to analyse the data to:

- Identify distinct customer segments based on their spending habits and previous interactions with the bank by employing clustering algorithms.
- Offer recommendations to the bank on improving marketing strategies and customer service tailored to these segments.

3. Data description

The data provided is of various customers of the bank and their financial attributes like credit limit, the total number of credit cards the customer has and different channels through which customers have contacted the bank for any queries (including visiting the bank, online and through a call centre).

The detailed data dictionary is given below.

- **SI_No:** Primary key of the records.
- **Customer Key:** Customer identification number.
- **Average_Credit_Limit:** Average credit limit of each customer for all credit cards.
- **Total_Credit_cards:** Total number of credit cards possessed by the customer.
- **Total_visits_bank:** Total number of visits that the customer made (yearly) personally to the bank.
- **Total_visits_online:** Total number of visits or online logins made by the customer (yearly).
- **Total_calls_made:** Total number of calls made by the customer to the bank or its customer service department (yearly).

4. Exploratory Data Analysis

4.1. Data structure

After importing **NumPy** and **Pandas** libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

	SI_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	1	87073	100000	2	1	1	0
1	2	38414	50000	3	0	10	9
2	3	17341	50000	7	1	3	4
3	4	40496	30000	5	1	1	4
4	5	47437	100000	6	0	12	3
5	6	58634	20000	3	0	1	8
6	7	48370	100000	5	0	11	2
7	8	37376	15000	3	0	1	1
8	9	82490	5000	2	0	2	2
9	10	44770	3000	4	0	1	7

Table 1. Data structure

OBSERVATION	VARIABLES
660	7

4.2. Data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 660 entries, 0 to 659
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   SI_No                 660 non-null   int64
1   Customer Key          660 non-null   int64
2   Avg_Credit_Limit      660 non-null   int64
3   Total_Credit_Cards    660 non-null   int64
4   Total_visits_bank     660 non-null   int64
5   Total_visits_online   660 non-null   int64
6   Total_calls_made      660 non-null   int64
dtypes: int64(7)
memory usage: 36.2 KB
```

Table 2. Data types

- The data set contains **seven variables** indexed from 0-6.
- The data set consists of 7 numerical columns of integer data type.
- The **SI_No** column and **Customer Key** column does not add value to the data. Hence can be dropped.

4.3. Null values

- No null values were found in the data set.

4.4. Duplicate Values

- No duplicate values were found in the data set.

4.5. Statistical Summary

	count	mean	std	min	25%	50%	75%	max
SI_No	660.0	330.500000	190.669872	1.0	165.75	330.5	495.25	660.0
Customer Key	660.0	55141.443939	25627.772200	11265.0	33825.25	53874.5	77202.50	99843.0
Avg_Credit_Limit	660.0	34574.242424	37625.487804	3000.0	10000.00	18000.0	48000.00	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.00	5.0	6.00	10.0
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.00	2.0	4.00	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.00	2.0	4.00	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.00	3.0	5.00	10.0

Table 3. Statistical summary

The following observations were made:

- **Average_Credit_Limit:** The average credit limit per customer varies between 3,000 and 2,00,000, with the mean credit limit approximately at 34,500. 75% of customers have an average limit of 48,000, indicating the potential presence of outliers.
- **Total_Credit_Cards:** The total number of credit cards owned by a customer range from 1 to 10, with an average of 5 credit cards. This indicates the potential presence of outliers, as the maximum is 10.
- **Total_visits_bank:** On average, customers have visited the bank twice, with a maximum of 5 visits in a year. Additionally, some customers have not visited the bank at all.
- **Total_visits_online:** On average, customers have visited the online website twice, with a maximum of 15 visits. Furthermore, some customers have not accessed the online website at all. Outliers could be present.
- **Total_calls_made:** The maximum number of calls made by customers annually is 10, with an average of 3 calls.

4.6. Count and Percentage of Variables in the Data Set

Total_Credit_Cards 4 151 6 117 7 101 5 74 2 64 1 59 3 53 10 19 9 11 8 11 Name: count, dtype: int64	Total_Credit_Cards 4 0.228788 6 0.177273 7 0.153030 5 0.112121 2 0.096970 1 0.089394 3 0.080303 10 0.028788 9 0.016667 8 0.016667 Name: proportion, dtype: float64
Total_visits_bank 2 158 1 112 0 100 3 100 5 98 4 92 Name: count, dtype: int64	Total_visits_bank 2 0.239394 1 0.169697 0 0.151515 3 0.151515 5 0.148485 4 0.139394 Name: proportion, dtype: float64
Total_visits_online 2 189 0 144 1 109 4 69 5 54 3 44 15 10 7 7 10 6 12 6 8 6 11 5 13 5 9 4 14 1 6 1 Name: count, dtype: int64	Total_visits_online 2 0.286364 0 0.218182 1 0.165152 4 0.104545 5 0.081818 3 0.066667 15 0.015152 7 0.010606 10 0.009091 12 0.009091 8 0.009091 11 0.007576 13 0.007576 9 0.006061 14 0.001515 6 0.001515 Name: proportion, dtype: float64
Total_calls_made 4 108 0 97 2 91 1 90 3 83 6 39 7 35 9 32 8 30 5 29 10 26 Name: count, dtype: int64	Total_calls_made 4 0.163636 0 0.146970 2 0.137879 1 0.136364 3 0.125758 6 0.059091 7 0.053030 9 0.048485 8 0.045455 5 0.043939 10 0.039394 Name: proportion, dtype: float64

Table 4. Count and percentage of variables

OBSERVATIONS:

- **Total_Credit_Cards:** Almost 30% of the customers own 4 credit cards, followed by 6 and 7 credit cards owned by 17% and 15% of the customers. 8 to 10 credit cards are owned by lesser than 2% customers.
- **Total_visits_bank:** Most customers have visited the bank twice annually which accounts to 24%. Almost 15% of the customers have not visited the bank at all.
- **Total_visits_online:** Most customers have accessed the online website annually, representing 28%, while nearly 21% of customers have not visited the site at all.
- **Total_calls_made:** 16% of the customers have made 4 calls and 4% constitutes 10 calls.

4.7. Univariate analysis

4.7.1. Average credit limit

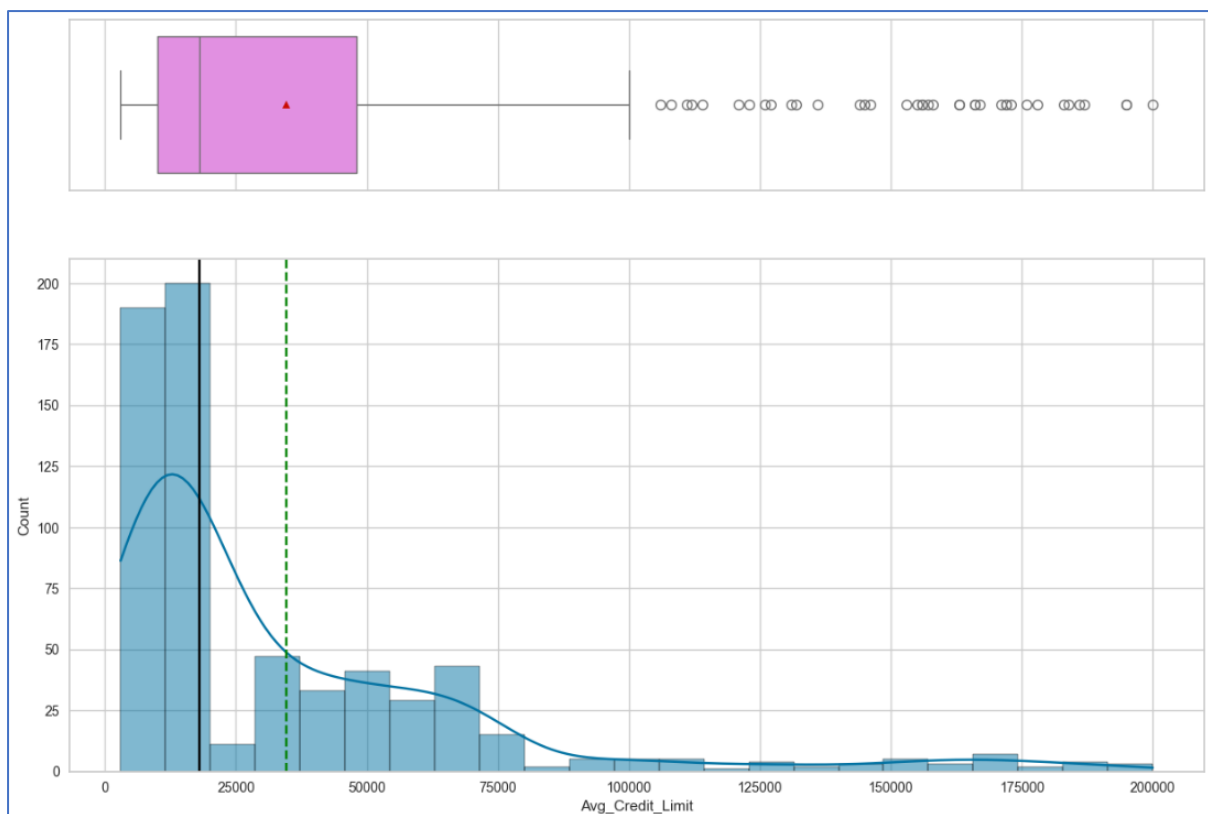


Fig 1. Histogram-Boxplot: Average credit limit

OBSERVATIONS:

- The average credit limit per customer is a highly right skewed distribution.
- The mean of the average credit limit per customer is close to 35,000.
- The outliers exceed 100,000, indicating that customers eligible for higher limits are few.
- The customers with average credit limit below 25,000 are high in number.

4.7.2. Total credit cards

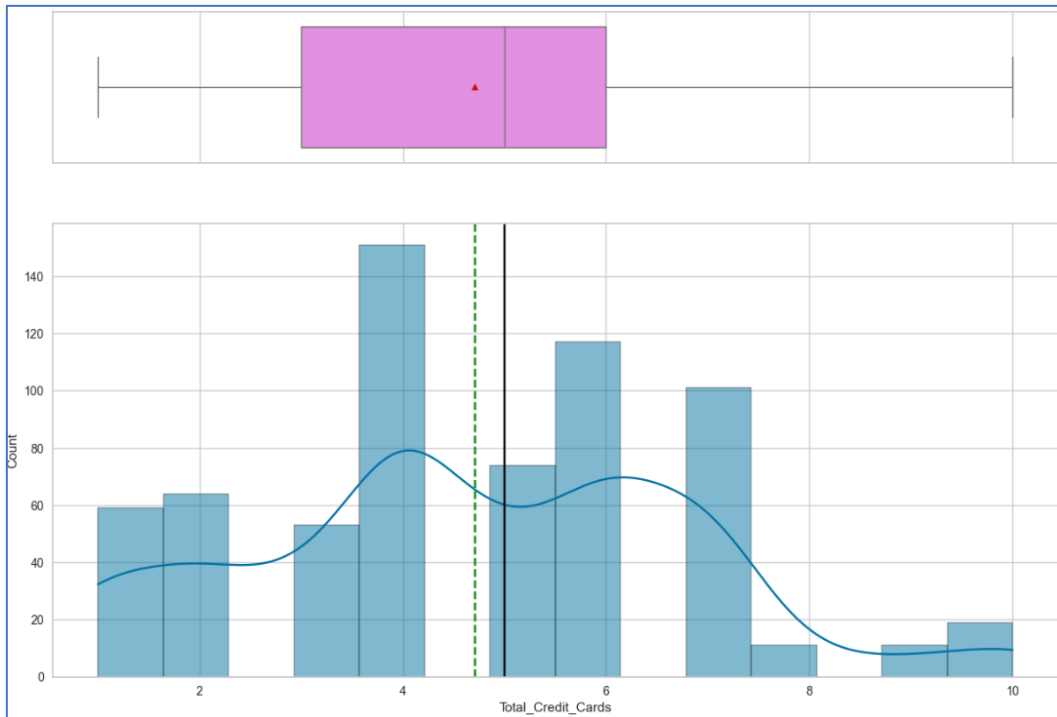


Fig 2. Histogram-Boxplot: Total credit cards

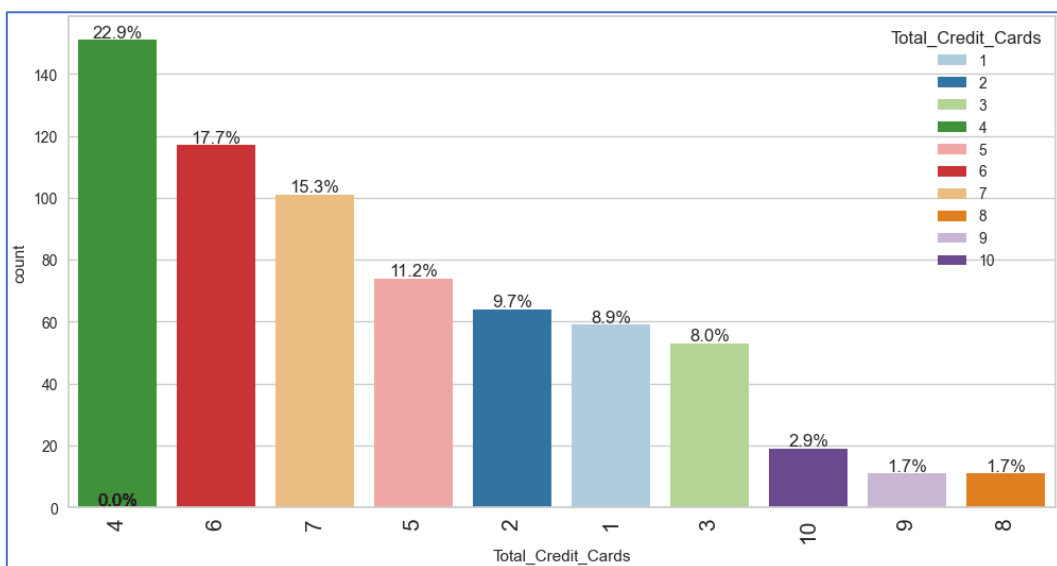


Fig 3. Barplot: Total credit cards

OBSERVATIONS:

- The total credit cards show a multimodal distribution.
- The number of customers with 4, 6 and 7 credit cards are comparatively higher with 23%, 17.7% and 15.3% respectively.
- The number of customers with 8 to 10 credit cards is relatively low with 1.7%, 1.7% and 2.9% respectively.

4.7.3. Total visits bank

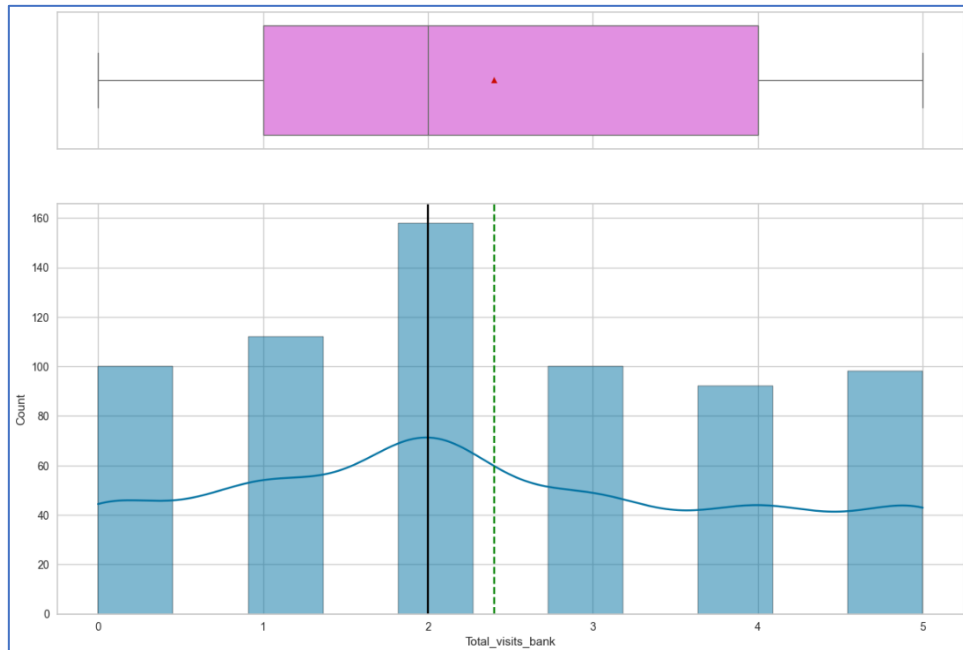


Fig 4. Histogram-Boxplot: Total visits bank

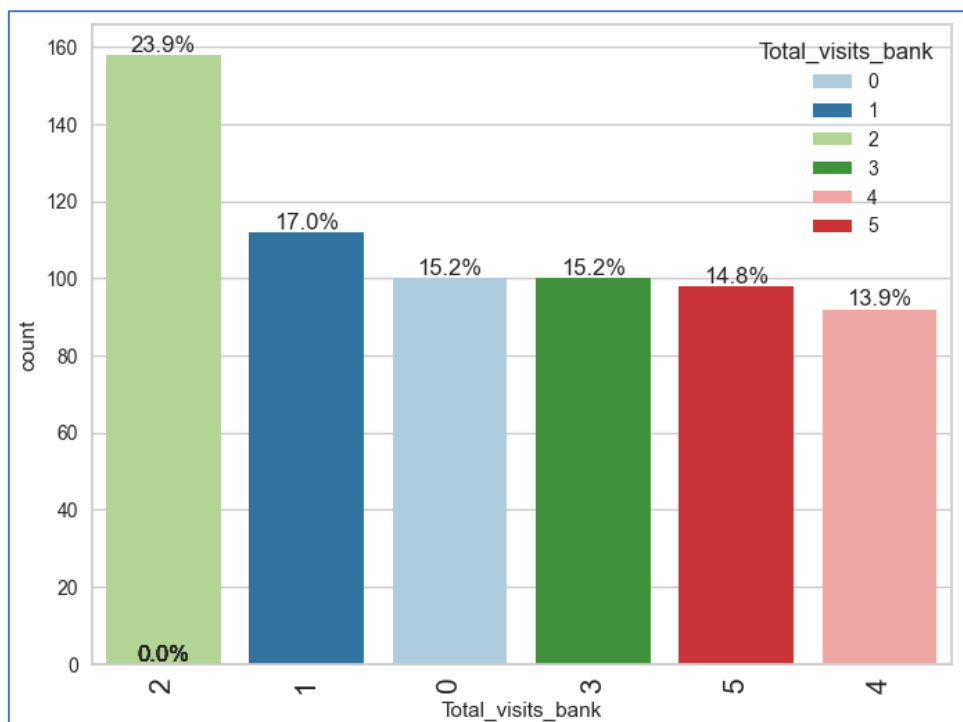


Fig 5. Barplot: Total visits bank

OBSERVATIONS:

- The total visits to the bank made by the customers show a multimodal distribution.
- The total number of visits made annually range from 0 to 5.
- 23% of customers have visited the bank twice.
- 15.2% of the customers have not visited the bank.

4.7.4. Total visits online

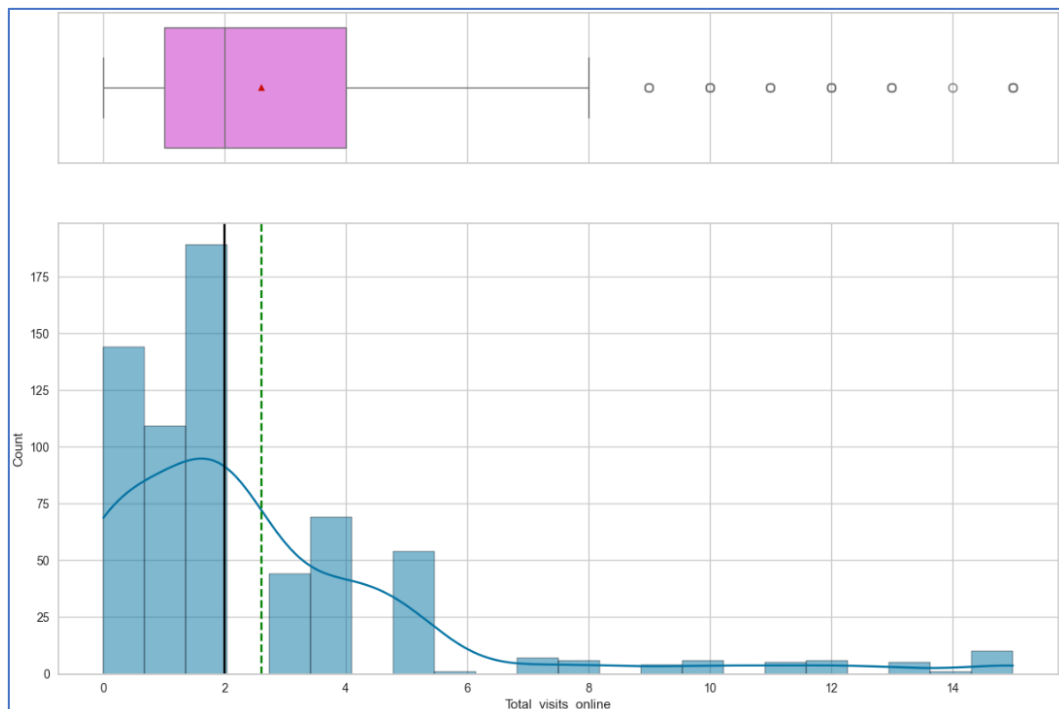


Fig 6. Histogram-Boxplot: Total visits online

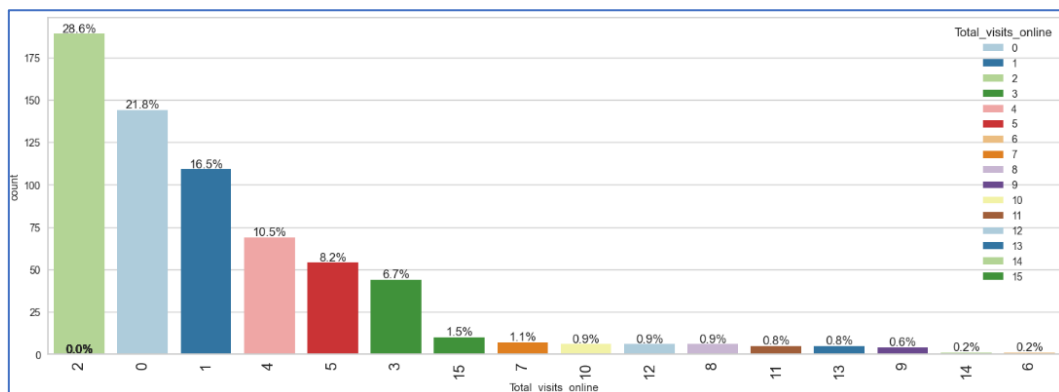


Fig 7. Barplot: Total visits online

OBSERVATIONS:

- The total visits to the online website made by the customers shows a highly right skewed distribution.
- The total number of visits made annually range from 1 to 15.
- 28.6% of customers have visited the online website twice and 21.8% of the customers have not visited at all.
- The number of customers visiting the online website more than 8 times is low, suggesting that only a few individuals are experiencing issues with credit card services.

4.7.5. Total calls made

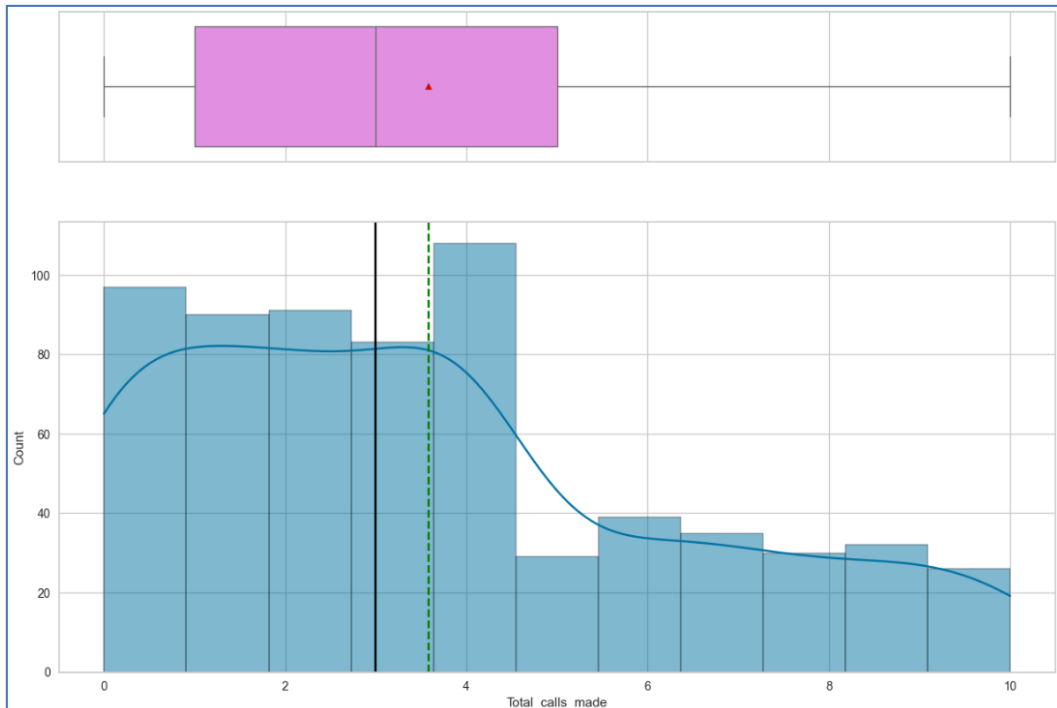


Fig 8. Histogram-Boxplot: Total calls made

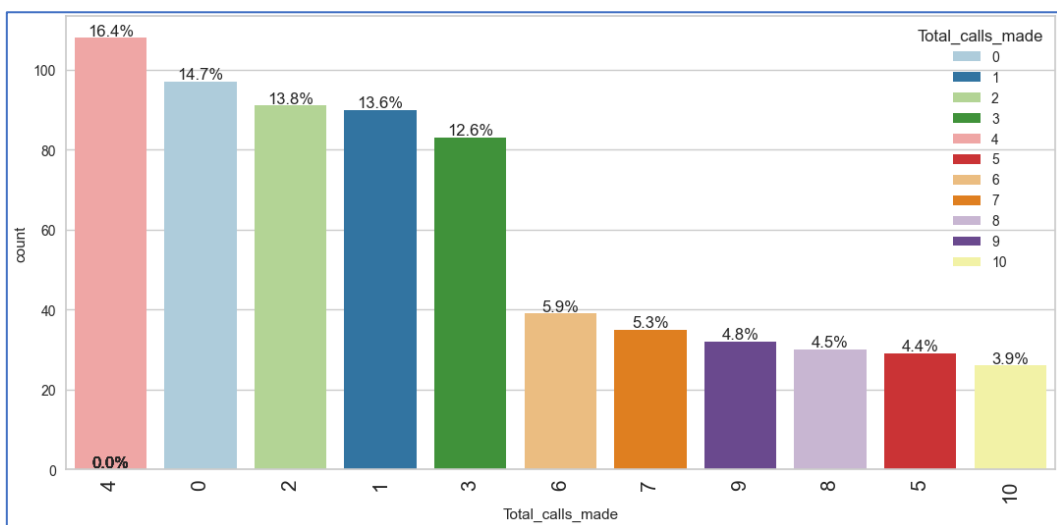


Fig 9. Barplot: Total calls made

OBSERVATIONS:

- The total calls made to the customer service department ranges from 0-10 showing a slightly right skewed distribution.
- The highest proportion of annual calls accounts for 16.4%, while approximately 14.7% of customers have not made any calls.
- The number of calls ranging from 0 to 4 is relatively high.

4.7.6. Cumulative Distribution Function plot

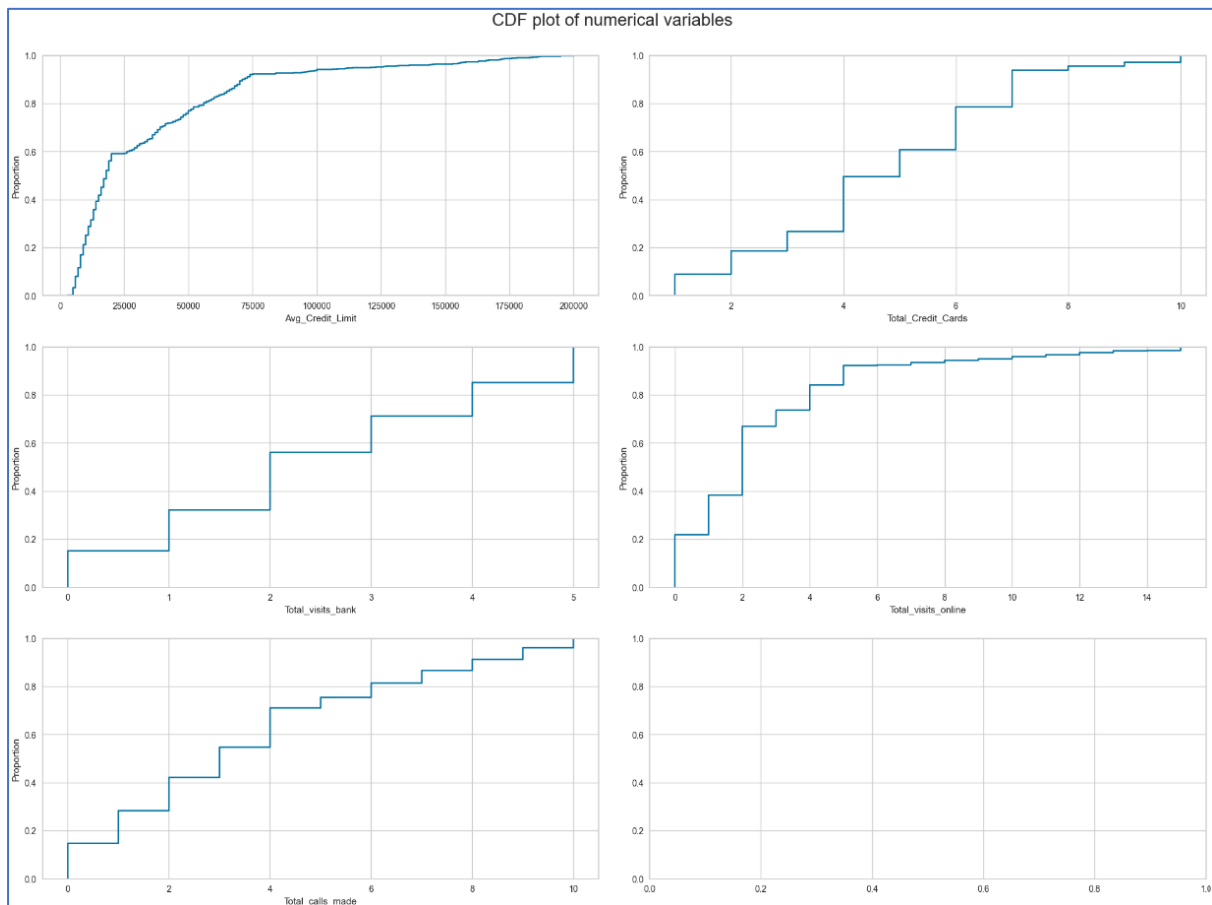


Fig 10. Cumulative Distribution Function plot

OBSERVATIONS:

- Nearly 95% of customers have an average credit limit below 75,000.
- 80% of customers own fewer than 6 credit cards.
- 70% of customers have visited the bank 4 times or fewer.
- 90% of customers have accessed the online website 8 times or less.
- 70% of customers have made 4 calls or fewer.

4.8 Bivariate analysis

4.8.1 Heat Map

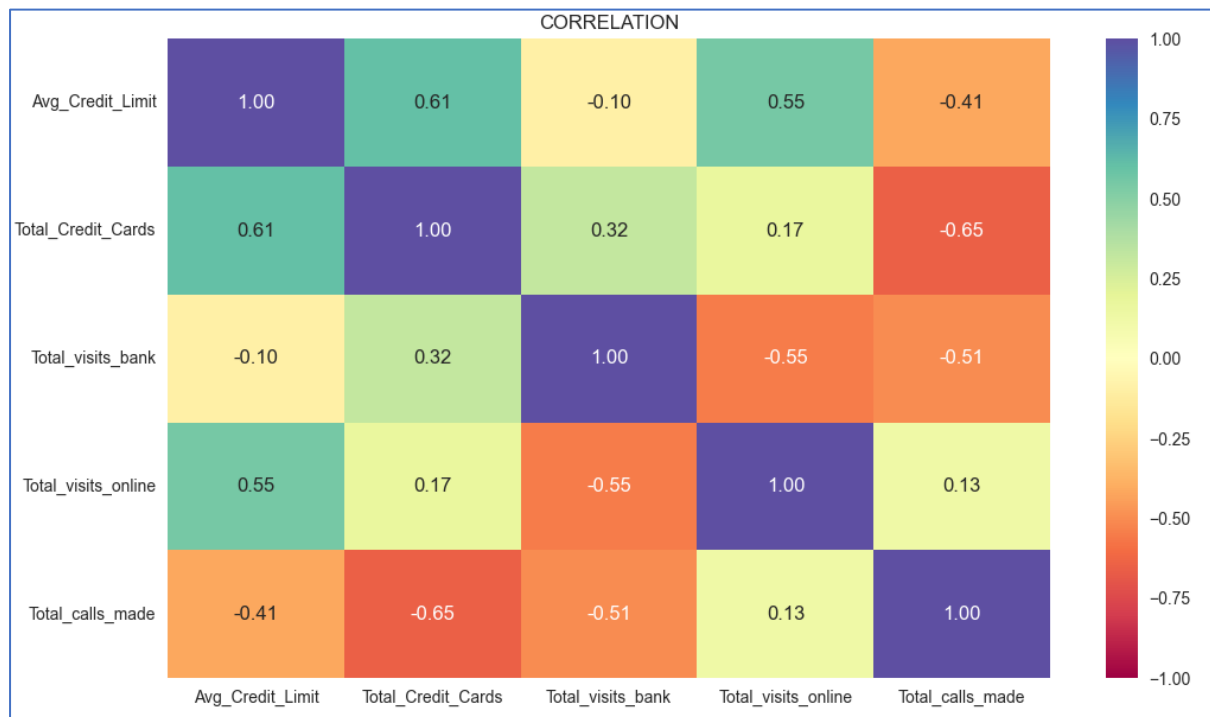


Fig 11. Heat map

OBSERVATIONS:

- There is a strong positive correlation (0.61) between the total number of credit cards owned and the average credit limit per customer.
- The total number of visits to the online website by customers is highly influenced by the average credit limit, with a correlation of 0.55.
- The total visits to the bank show a slight positive correlation (0.32) with the number of credit cards owned by the customer.
- As the number of credit cards owned increases, the total number of calls made decreases, indicating a negative correlation of -0.65.
- There is a negative correlation of -0.55 between total online visits and total bank visits, which makes sense.
- The total number of calls made decreases by 0.51 as the average credit limit and bank visits increase.

4.8.2 Pair Plot for understanding the relationship between numerical variables

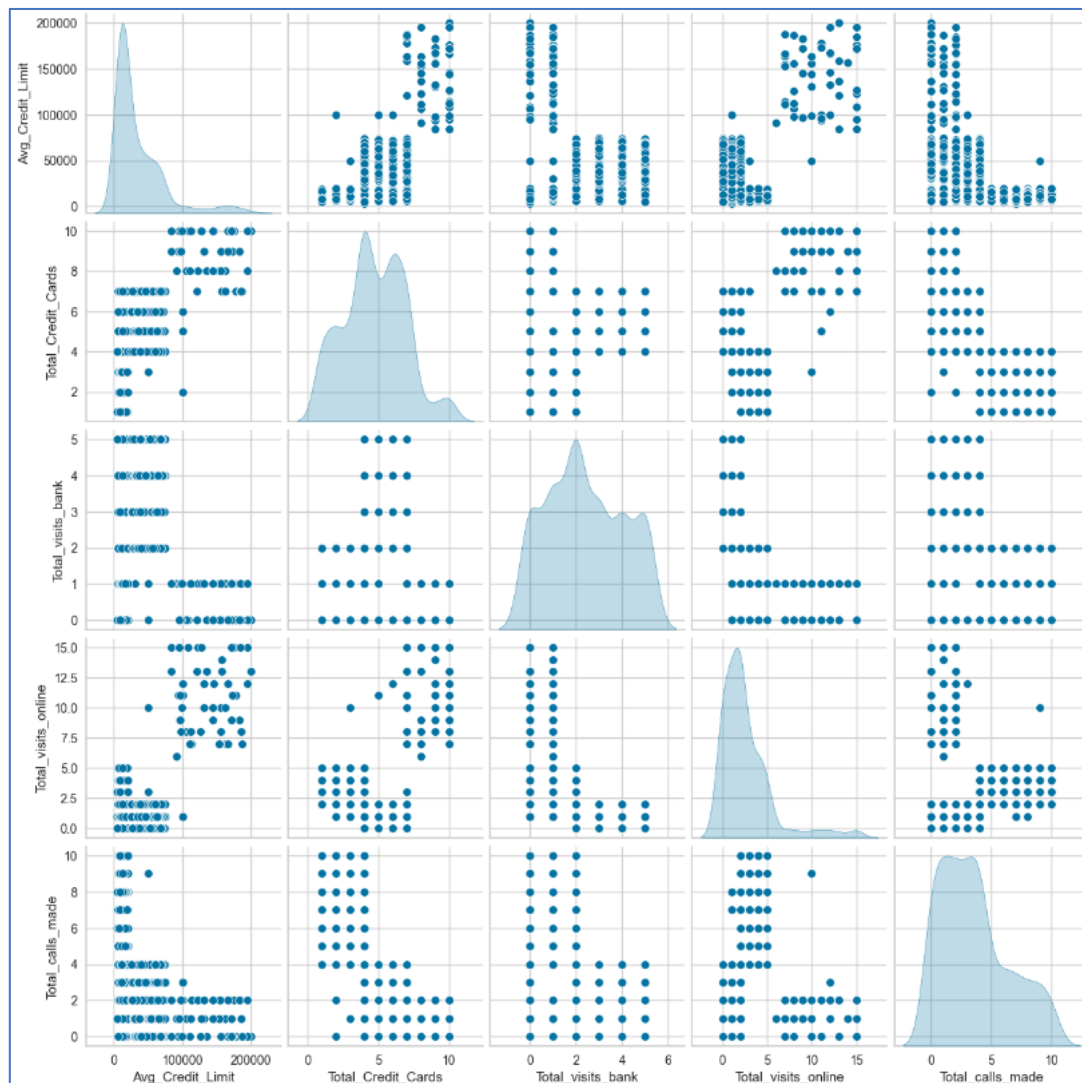


Fig 12. Pair plot

OBSERVATIONS:

- The average credit limit of each customer for all credit cards appears to have a positive correlation with several variables.
- Both the total number of visits to the online website and the total calls made show a strong negative correlation with total bank visits.
- The average credit limit, total visits online and total calls made show a right skewed plot.
- The pair plot provides insights into the potential number of clusters for K-means clustering, suggesting that a range of 4 to 6 clusters could be appropriate.

4.9 Insights based on Exploratory Data Analysis

4.9.1 Based on Univariate analysis

- **Avg_Credit_Limit:** The average credit limit per customer is highly right-skewed, with a mean of around 35,000, indicating that not all customers receive high limits. 95% of customers have an average credit limit below 75,000. This suggests that credit limits are determined by individual transaction behaviours.
- **Total_Credit_cards:** A significant percentage of customers, about 60% own fewer than 6 credit cards. In contrast, the relatively low percentages for customers with 8 to 10 credit cards potentially indicate that most customers prefer to maintain a manageable number of cards. This pattern suggests a tendency toward moderation in credit card ownership among the majority of customers.
- **Total_visits_bank:** The multimodal distribution of total bank visits suggests that customers cluster around specific visit frequencies. With a range of 0 to 5 visits annually, the data reveals that 23% of customers visit the bank twice, indicating this is a common behaviour. The patterns imply that while some customers engage with the bank regularly, a notable minority do not visit at all, highlighting differing preferences in banking habits.
- **Total_visits_online:** 90% of customers have accessed the online website 8 times or less. The low number of customers visiting the website more than 8 times implies that only a few individuals may be facing issues with credit card services, as frequent visits could indicate a need for assistance or support.
- **Total_calls_made:** The highest proportion of calls (16.4%) suggests that a small group of customers is more active in seeking assistance, while approximately 14.7% have not made any calls at all. The relatively high number of calls ranging from 0 to 4 implies that many customers either have minimal service needs or are able to resolve issues without contacting customer service. Overall, this pattern suggests a general trend of limited reliance on customer service among the majority of customers.

4.9.2 Based on Bivariate analysis

- **Credit Cards and Credit Limits:** The strong positive correlation (0.61) between the total number of credit cards owned and the average credit limit indicates that customers with more credit cards tend to have higher credit limits. This suggests that financial institutions may be more willing to extend higher limits to customers who already have multiple cards, possibly due to their demonstrated creditworthiness.
- **Credit Limits and Online Engagement:** The correlation (0.55) between average credit limit and total online visits suggests that customers with higher credit limits are more engaged with online banking or services. This might imply that these customers are more financially active or prefer online management of their accounts.

- **Credit Cards and Bank Visits:** The slight positive correlation (0.32) between total bank visits and the number of credit cards suggests that while there is some relationship, it is not strong. Customers with more credit cards may still visit the bank, but other factors might be influencing their in-person visits.
- **Credit Cards and Calls Made:** The negative correlation (-0.65) between the number of credit cards owned and the total number of calls made implies that customers who own more credit cards are likely to require less assistance or support. This may indicate a greater level of comfort with their financial management, possibly due to experience.
- **Online Visits and Bank Visits:** The negative correlation (-0.55) between total online visits and total bank visits reinforces the idea that as customers engage more online, they tend to visit physical bank locations less. This aligns with broader trends in banking, where online services reduce the need for in-person visits.
- **Calls and Engagement:** The statement that the total number of calls decreases by 0.51 as both the average credit limit and bank visits increase suggests that as customers engage more through online platforms (with higher credit limits) and visit banks less, they may also have fewer questions or need less assistance.
- **Skewness of Variables:** The right skewed distribution of average credit limit, total online visits, and total calls suggests that while most customers may have lower values, there are a few customers with significantly higher values. This could imply a concentration of high-value customers with greater financial engagement.

5. Data Preprocessing

5.1. Duplicate value check

- No duplicate values are found in the data set.

5.2. Missing value treatment

- No missing values or null values are found in the data set.

5.3. Outlier detection

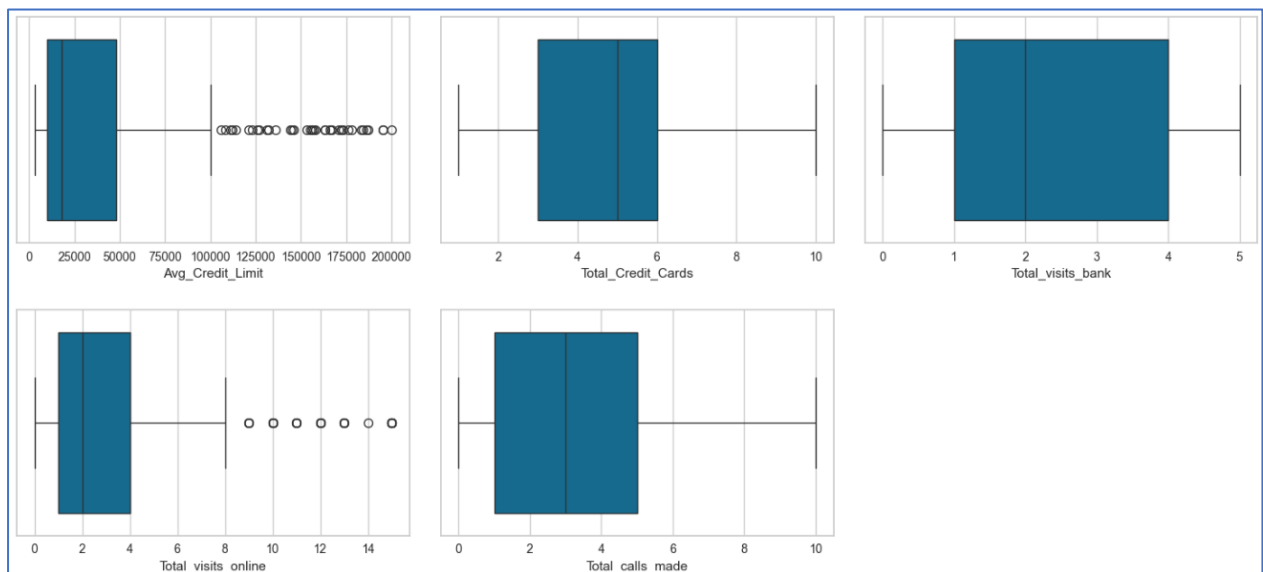


Fig 13. Outlier detection

OBSERVATIONS:

- There are outliers in the **Average_Credit_Limit** and **Total_visits_online** column.
- These outliers should not be removed, as they provide valuable insights. The high average credit limit for eligible customers may reflect their strong transaction history, while the total online visits could indicate concerns regarding the service. Further analysis is necessary to better understand these trends.
- Thereby, we will not treat them.

5.4. Feature Engineering

- The **SI_No** column and **Customer Key** column are dropped from the data set as they do not add value to the analysis.

5.5. Data Scaling

- The dataset has been scaled using **StandardScaler** from **sklearn.preprocessing**.

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	1.740187	-1.249225	-0.860451	-0.547490	-1.251537
1	0.410293	-0.787585	-1.473731	2.520519	1.891859
2	0.410293	1.058973	-0.860451	0.134290	0.145528
3	-0.121665	0.135694	-0.860451	-0.547490	0.145528
4	1.740187	0.597334	-1.473731	3.202298	-0.203739

Table 5. Scaled data

6. Model Building

Clustering is a powerful technique for uncovering patterns in data. By following a structured approach to model building, we can derive meaningful insights from our dataset. Understanding our data, choosing the right algorithm, and validating the clusters are crucial steps in building an effective clustering model.

6.1. K-means Clustering

K-means clustering is a versatile and widely used technique in data analysis and machine learning. It is a popular unsupervised machine learning algorithm used to partition a dataset into distinct groups (clusters).

6.1.1. Determining the number of clusters (k)

Choosing the right number of clusters is crucial for effective clustering. Common methods include:

- **Elbow Method:** Plotting the distortion score against different values of k to find the "elbow" point where the rate of improvement slows.
- **Silhouette Score:** Measuring how similar each point is to its own cluster compared to other clusters using `silhouette_score` algorithm from `sklearn.metrics`. Higher scores indicate better-defined clusters.]

6.1.2. Elbow method

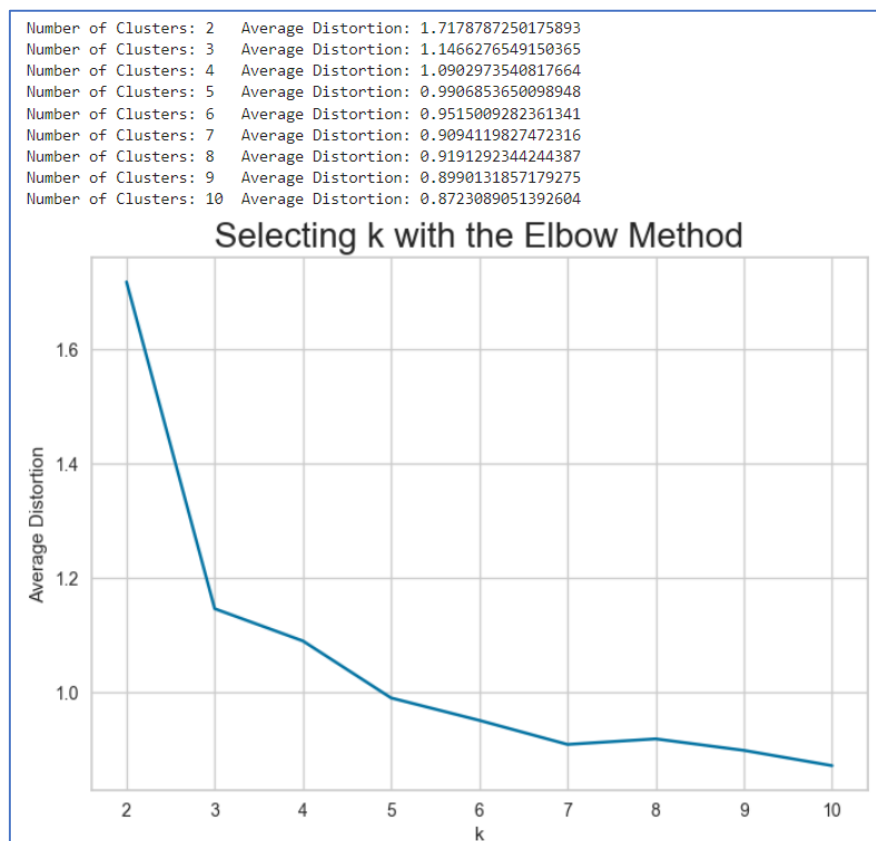


Fig 14. Elbow method

OBSERVATIONS:

- The elbow point is where the rate of decrease in distortion starts to slow down significantly. This indicates that adding more clusters yields diminishing returns in terms of improving the fit.
- When interpreting the above graph, it can be seen that 3, 4, or 5 could be optimal numbers of clusters which indicates that the distortion scores for these values show favourable characteristics.
- Number of clusters (k) = 3: This option provides a straightforward model that is easy to interpret and can capture major patterns in the data without overfitting.
- Number of clusters (k) = 4: This gives a bit more granularity than 3, potentially capturing more nuanced patterns or segments in the data.
- Number of clusters (k) = 5: This offers even more detail, but we must assess whether the additional complexity is meaningful. It might capture slight variations in the data.

To enhance our analysis, we should examine the computation time needed for each number of clusters.

The fit time on the elbow graph helps assess the computational efficiency of clustering while also aiding in making informed decisions about the optimal number of clusters to use, balancing model complexity and performance.

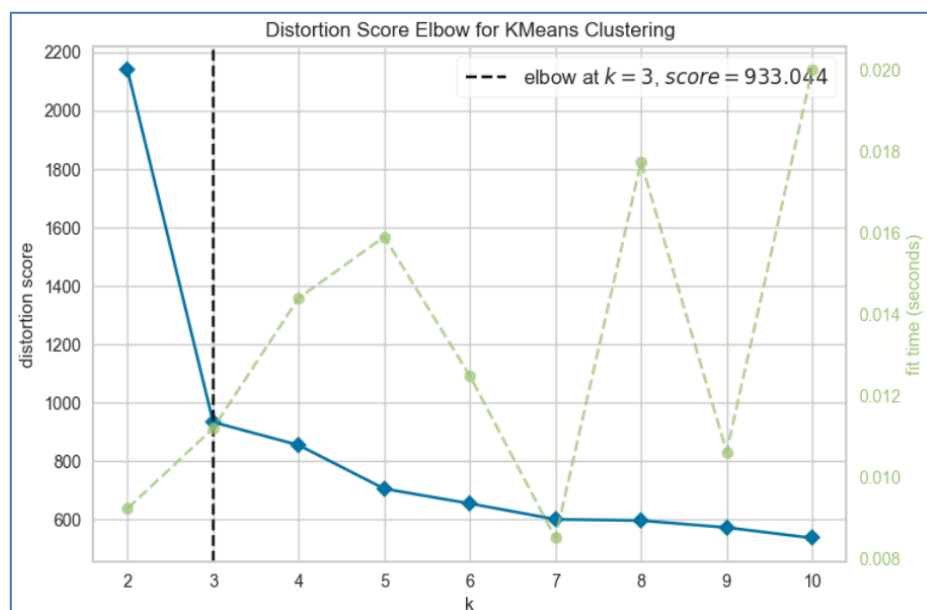


Fig 15. Elbow method with computational time

OBSERVATIONS:

- Among the options of 3, 4, and 5 clusters, the computation time for 5 clusters is the highest at 0.016 seconds, followed by 4 clusters at 0.014 seconds, while 3 clusters have the lowest computation time at 0.011 seconds.
- As the value of k increases, the fit time typically increases, since the algorithm needs to perform more calculations to assign points to more clusters. The computational time for 3 clusters seems to be lesser.

6.1.3. Silhouette Score

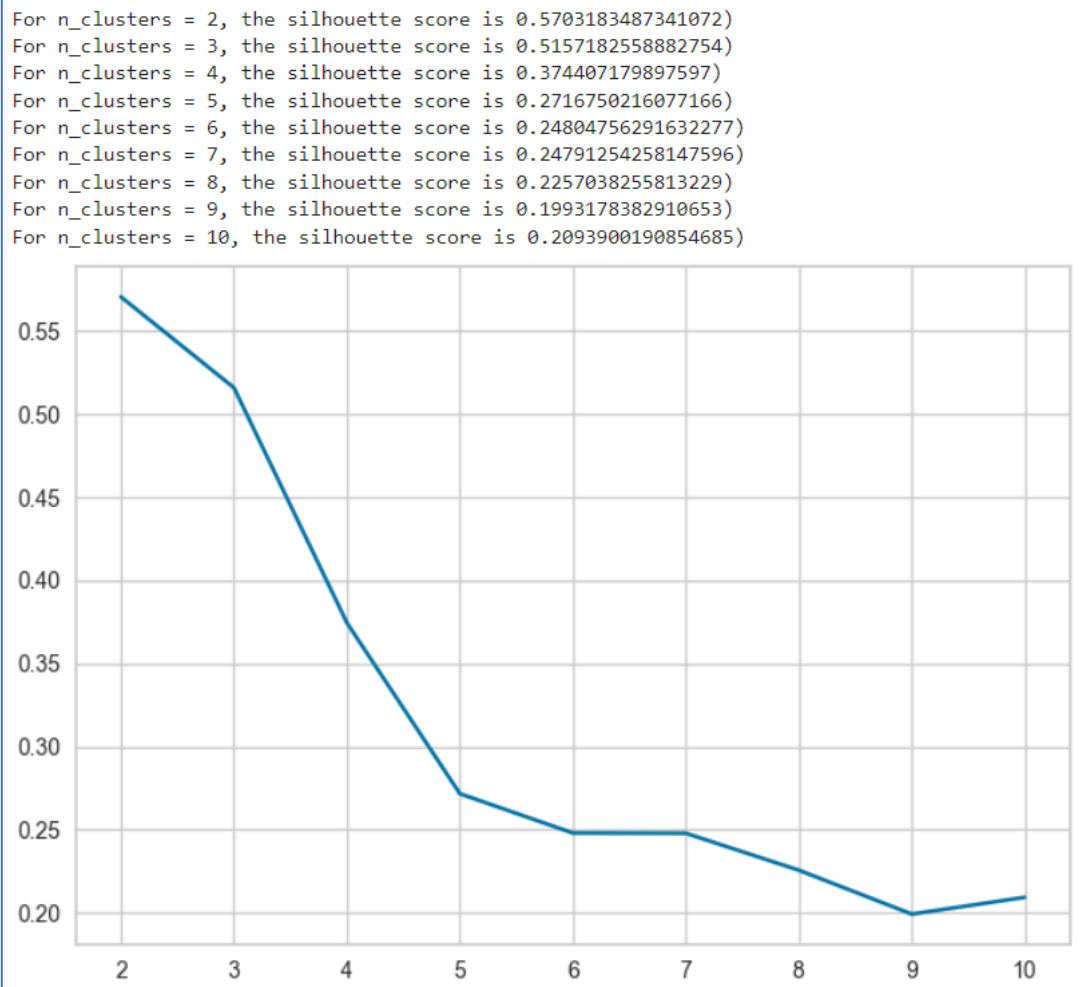


Fig 16. Silhouette score

OBSERVATIONS:

- The silhouette score is highest for k=3, reaching 0.515, followed by scores of 0.374 for k=4 and 0.271 for k=5.
- As the value of k increases, the silhouette score tends to decrease. A lower score indicates that points may be incorrectly clustered or that clusters are overlapping.
- Therefore, it is crucial to select a value of k that yields a high silhouette score.

To enhance our analysis, we should examine the computation time needed for each number of clusters.

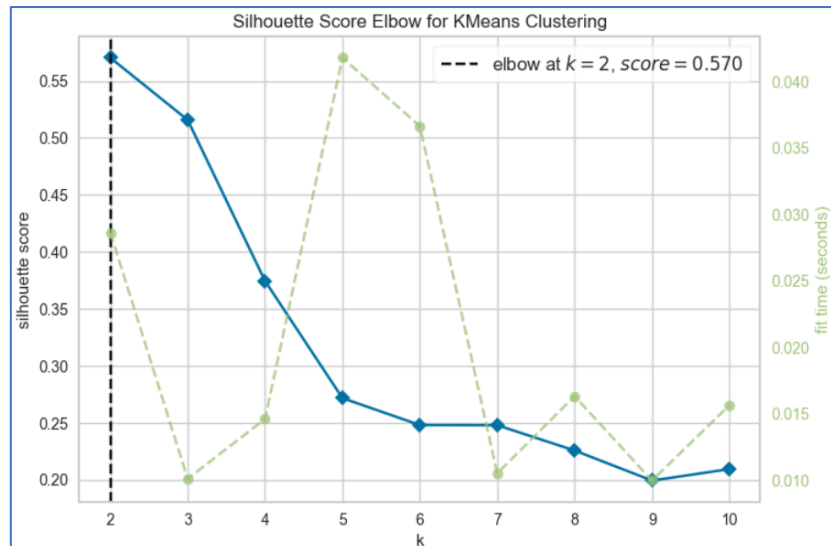


Fig 17. Silhouette score with computational time

OBSERVATIONS:

- The computation time is highest for 5 clusters at 0.040 seconds, followed by 4 clusters at 0.015 seconds, while 3 clusters have the lowest computation time at 0.010 seconds.
- Given these observations, it would be prudent to eliminate k=5 due to its significantly higher computation time, and consider either k=3 or k=4 since their computation times are nearly equivalent.

Let us consider silhouette plot for values k=2, k=3, k=4, k=5 and k=6 to obtain a visual representation of how well each data point is clustered. The overall shape of the silhouette plot can help assess the quality of the clustering.

Silhouette plot: k=2

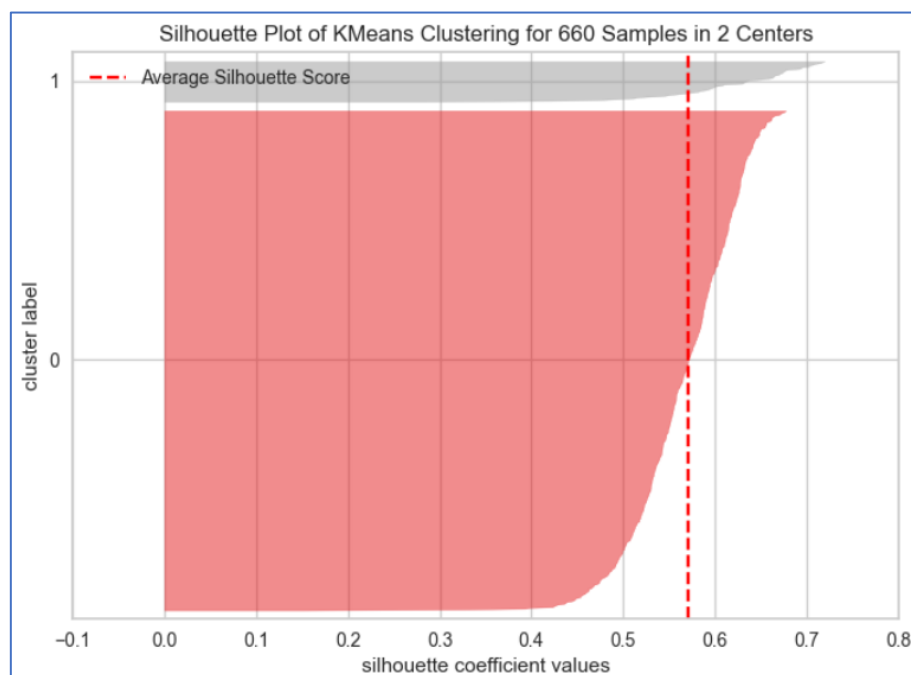
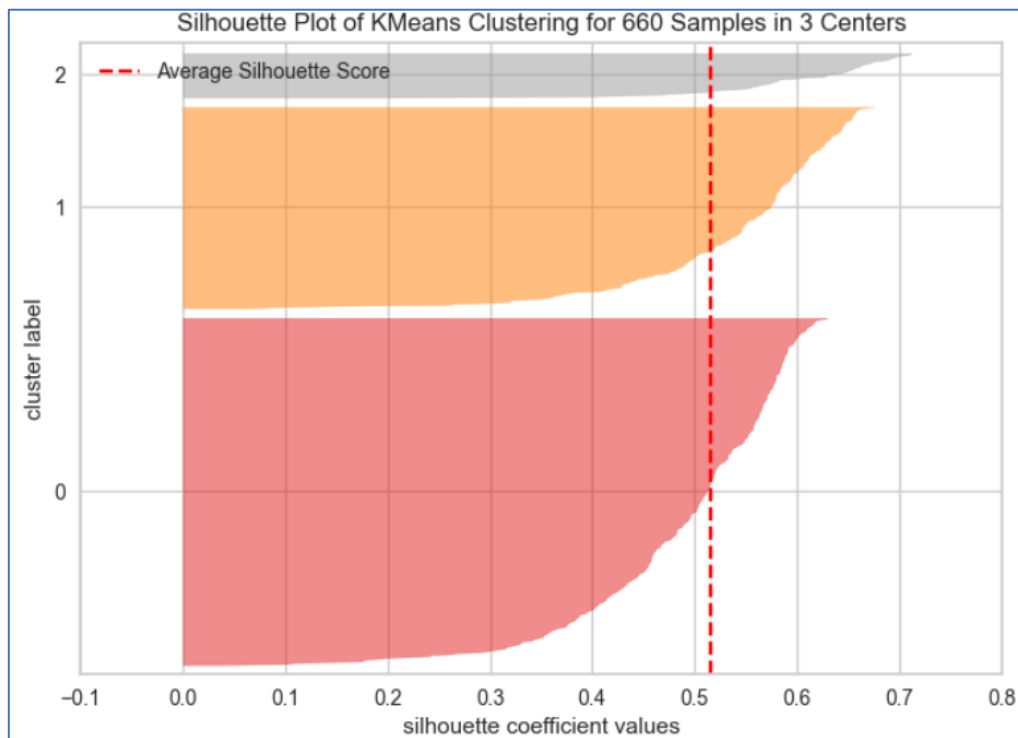
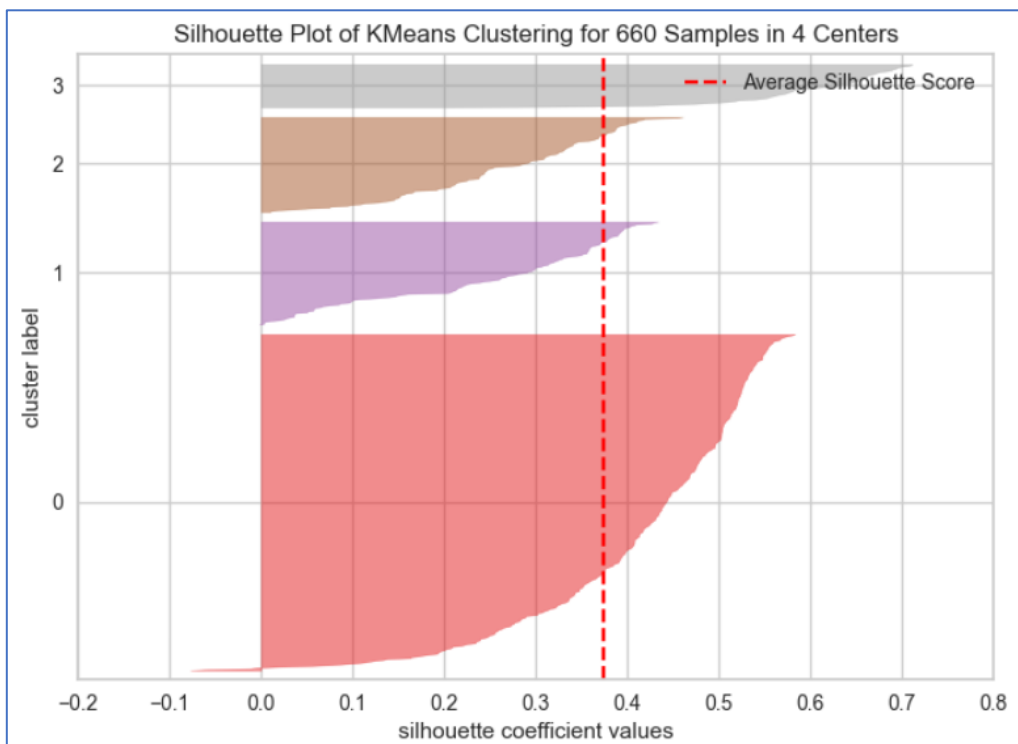


Fig 18. Silhouette Plot (k=2)

Silhouette plot: k=3**Fig 19. Silhouette Plot (k=3)****Silhouette plot: k=4****Fig 20. Silhouette Plot (k=4)**

Silhouette plot: k=5

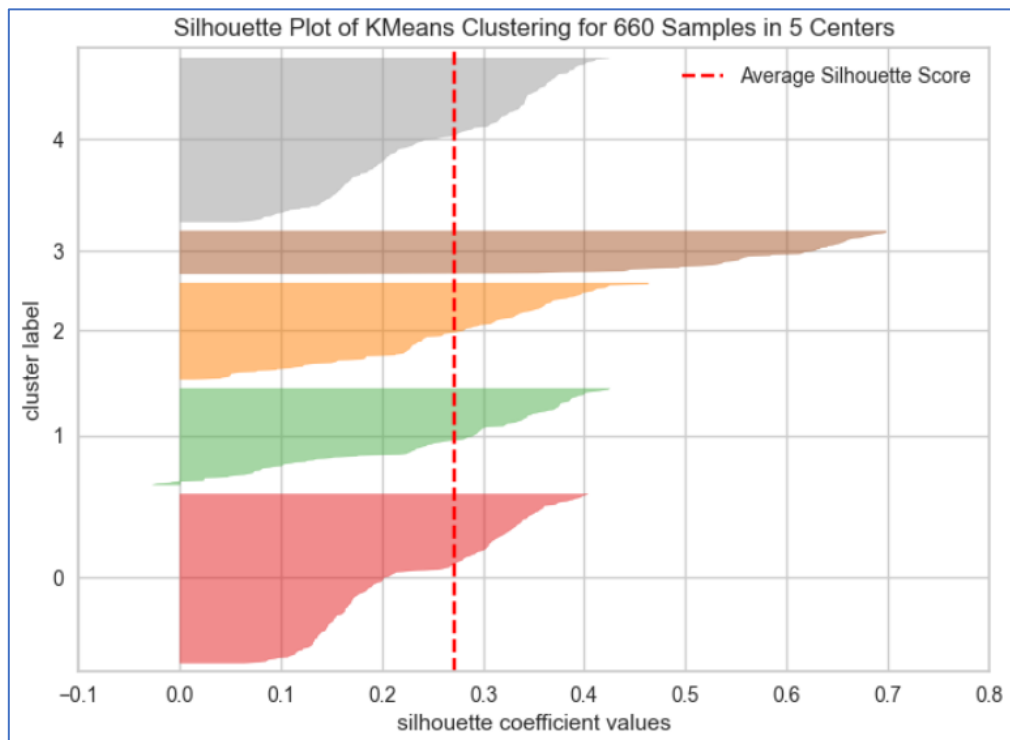


Fig 21. Silhouette Plot (k=5)

Silhouette plot: k=6

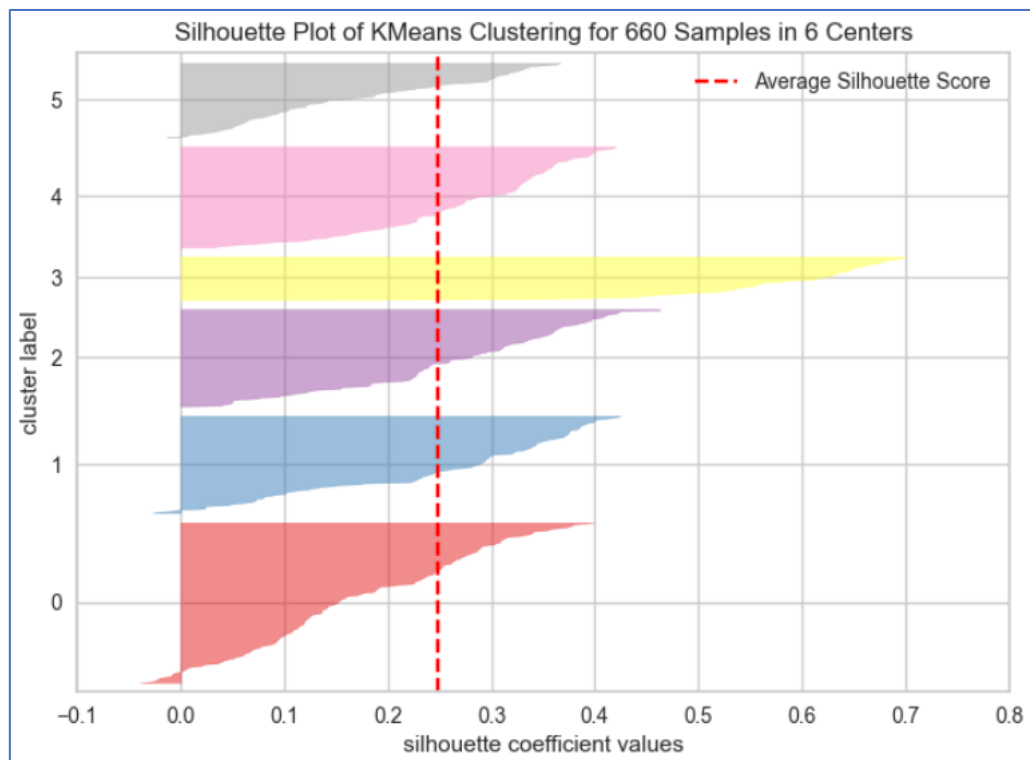


Fig 22. Silhouette Plot (k=6)

OBSERVATIONS:

- Based on the silhouette plots for the dataset, k=3 shows a good silhouette score of 0.515 with a low computation time of 0.010 seconds, while k=4 has a silhouette score of 0.374 and a computation time of 0.015 seconds.
- Choosing too few clusters may oversimplify the data, obscuring important patterns and variations, which can result in the loss of valuable insights for analysis.
- Although a lower number of clusters can make analysis simpler, it often compromises depth and accuracy, making it more challenging to draw meaningful conclusions from the data.
- As the computation time of k=4 is quite comparable to k=3, and the silhouette score for k=5 shows a significant drop, we will proceed with **k=4** for further analysis.

6.1.4. Creating Final Model

With **k chosen as 4**, we will assign K-Means cluster labels to the original dataframe and create a final model by using the **KMeans** algorithm from **sklearn.cluster**.

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	KM_segments
0	100000	2	1	1	0	0
1	50000	3	0	10	9	1
2	50000	7	1	3	4	0
3	30000	5	1	1	4	2
4	100000	6	0	12	3	3

Table 6. K-means: Cluster label assigned original dataframe

6.1.5. Cluster profiling

Cluster profiling helps in understanding the characteristics of different groups (or clusters) within the dataset. This is done by grouping the data based on the clusters formed and checking the mean, median and boxplot of each variable with respect to the clusters.

In order to identify distinct customer segments based on their spending habits, the **Avg_Credit_Limit** variable is used for grouping.

6.1.5.1. Cluster Profile using mean

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segments
KM_segments						
0	31226.315789	5.478947	4.515789	1.010526	1.936842	190
1	35904.522613	5.522613	2.477387	0.969849	2.075377	199
2	141040.000000	8.740000	0.600000	10.900000	1.080000	50
3	12167.420814	2.393665	0.927602	3.574661	6.923077	221

Table 7. K-means: Cluster profiling using mean

6.1.5.2. Cluster Profile using median

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segments
KM_segments						
0	28000.000000	6.000000	5.000000	1.000000	2.000000	190
1	34000.000000	6.000000	3.000000	1.000000	2.000000	199
2	145500.000000	9.000000	1.000000	11.000000	1.000000	50
3	12000.000000	2.000000	1.000000	4.000000	7.000000	221

Table 8. K-means: Cluster profiling using median

6.1.5.3. Cluster Profile using boxplot

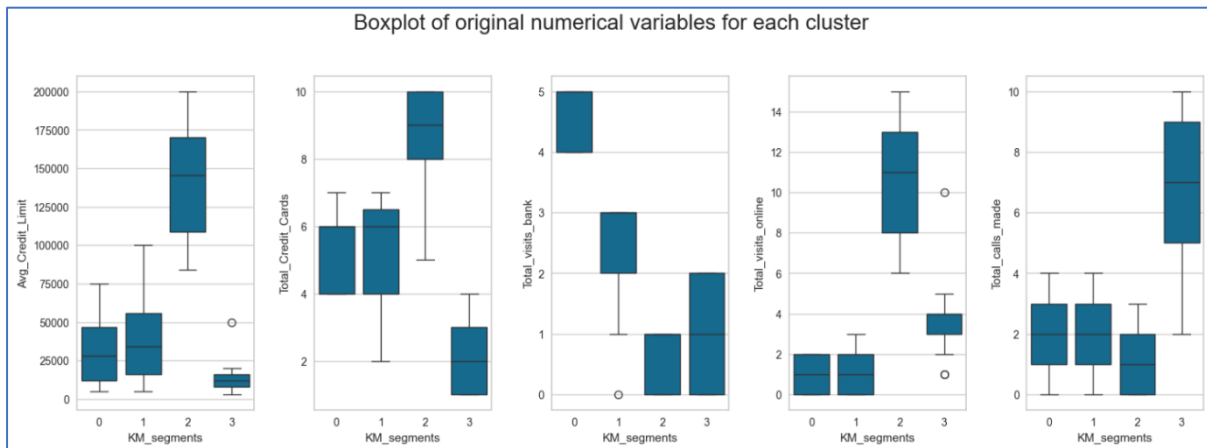


Fig 23. K-means: Cluster profiling using boxplot

OBSERVATIONS:

Cluster 0

- This cluster includes about 190 customers, almost 28% of the customers having credit cards.
- This cluster includes customers with an average credit limit of approximately 28,000 to 31,000, as determined by median and mean cluster profiling. The average credit limit suggests these customers are likely to have a good credit history, making them valuable to the bank.
- Customers typically own 6 credit cards on an average. They possibly use multiple cards for different purposes.
- They have visited the bank 5 times, but their online engagement is quite low, with only 1 online visit. Frequent bank visits suggest that these customers prefer face-to-face interactions, which could point to a lack of confidence in online banking or a preference for personal service.
- Additionally, they make just 2 annual calls to the bank. It could suggest satisfaction with current services or a lack of awareness about additional offerings.

Cluster 1

- This cluster consists of approximately 199 customers, representing nearly 30% of those with credit cards.
- Customers in this group have an average credit limit of around 35,000, and most credit card holders belong to this cluster. These customers are likely financially stable and trusted by the bank, making them valuable clients.
- On average, these customers own 6 credit card which indicates a tendency toward diverse financial products.
- They have visited the bank 3 times, while their online engagement is low, with only 1 online visit. This suggests that these customers may not be fully utilizing their cards.
- Furthermore, they make just 2 annual calls to the bank, which may indicate lack of awareness regarding additional offerings.

Cluster 2

- This cluster includes only 50 customers. They have an average credit limit of 141,0500, indicating that these individuals are highly eligible and experienced with credit cards. They are highly eligible for credit and have a strong financial background, suggesting they are considered low-risk by the bank.
- On average, they possess the highest number of credit cards among all clusters, nearly 9. This indicated their experience with managing credit and desire to leverage different rewards and benefits.
- They have made only 1 bank visit and 1 call, suggesting they do not require assistance.
- However, their online activity is notably high, with about 11 visits, likely for online transactions and bill payments.

Cluster 3

- This cluster includes approximately 221 customers, making up nearly 33% of all credit card holders, which is the highest percentage.
- The average credit limit for these customers is 12,000, the lowest among the clusters. This indicates that these customers are less experienced or have lower creditworthiness which could imply a need for financial education or support in managing credit.
- They have made a total of 1 bank visit and 4 online visits. These customers may prefer self-service options. However, the low number of in-person interactions could also suggest they are not fully utilizing available banking services.
- Additionally, they have the highest number of calls made, which is 7. This could indicate a need for assistance or clarification on banking services, which could reflect confusion about their credit options.

6.2 Hierarchical Clustering

Agglomerative hierarchical clustering is used for grouping the similar objects in the data set into clusters through a bottom-up approach.

6.2.1 Computing Cophenetic Correlation

Cophenetic Correlation evaluates how well the clustering structure reflects the original data. It compares the cophenetic distances derived from the hierarchical clustering with the original pairwise distances between the data points.

In order to find the cophenetic correlation, we employ the distance metrics selected from Euclidean, Chebyshev, Mahalanobis and Cityblock (also referred to as Manhattan distance) and linkage methods selected from single, complete, average and weighted.

The results are as follows:

```
Cophenetic correlation for Euclidean distance and single linkage is 0.7391220243806552.
Cophenetic correlation for Euclidean distance and complete linkage is 0.8599730607972423.
Cophenetic correlation for Euclidean distance and average linkage is 0.8977080867389372.
Cophenetic correlation for Euclidean distance and weighted linkage is 0.8861746814895477.
Cophenetic correlation for Chebyshev distance and single linkage is 0.7382354769296767.
Cophenetic correlation for Chebyshev distance and complete linkage is 0.8533474836336782.
Cophenetic correlation for Chebyshev distance and average linkage is 0.8974159511838106.
Cophenetic correlation for Chebyshev distance and weighted linkage is 0.8913624010768603.
Cophenetic correlation for Mahalanobis distance and single linkage is 0.7058064784553605.
Cophenetic correlation for Mahalanobis distance and complete linkage is 0.666353446387536.
Cophenetic correlation for Mahalanobis distance and average linkage is 0.8326994115042136.
Cophenetic correlation for Mahalanobis distance and weighted linkage is 0.7805990615142517.
Cophenetic correlation for Cityblock distance and single linkage is 0.7252379350252723.
Cophenetic correlation for Cityblock distance and complete linkage is 0.8731477899179829.
Cophenetic correlation for Cityblock distance and average linkage is 0.896329431104133.
Cophenetic correlation for Cityblock distance and weighted linkage is 0.8825520731498188.
```

Table 9. Cophenetic correlation

The highest cophenetic correlation of 1 indicates a perfect match between the original distances and cophenetic distances, 0 indicates no correlation and negative values suggest a negative correlation.

```
Highest cophenetic correlation is 0.8977080867389372, which is obtained with Euclidean distance and average linkage.
```

Table 10. Highest cophenetic correlation

OBSERVATIONS

- We see that the cophenetic correlation of 0.897 is maximum with **Euclidean distance** and **average linkage**.

6.2.2 Checking Dendrograms

Dendrogram is a tree-like diagram which visually represents the arrangement of clusters formed during hierarchical clustering. It displays the relationships between data points or clusters based on their similarity or dissimilarity.

Using the **Euclidean distance** metric and linkage methods selected from single, complete, average, centroid, ward and weighted, dendrograms are obtained.

Dendrogram: Single Linkage

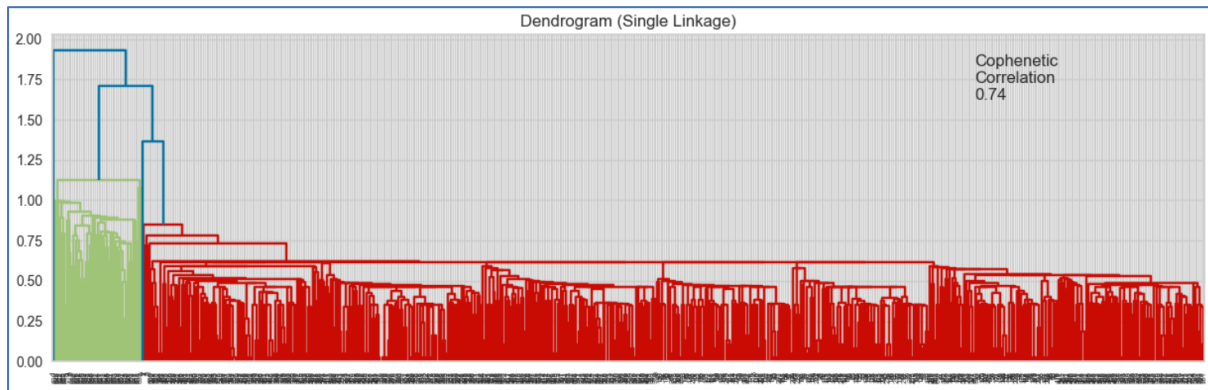


Fig 24. Dendrogram: Single Linkage

Dendrogram: Complete Linkage

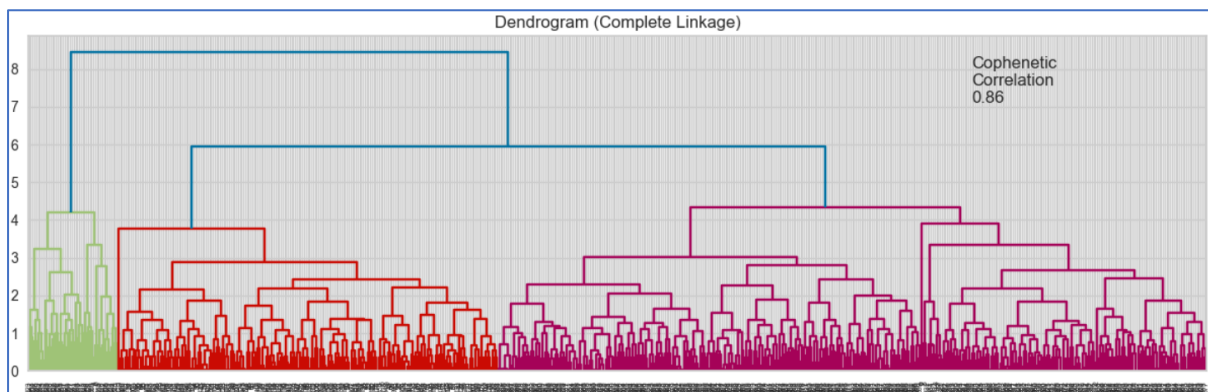


Fig 25. Dendrogram: Complete Linkage

Dendrogram: Average Linkage

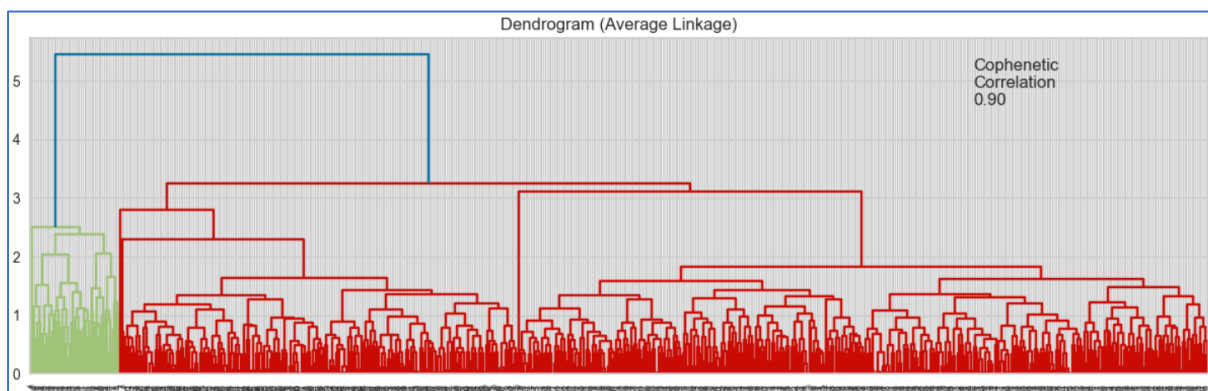


Fig 26. Dendrogram: Average Linkage

Dendrogram: Centroid Linkage

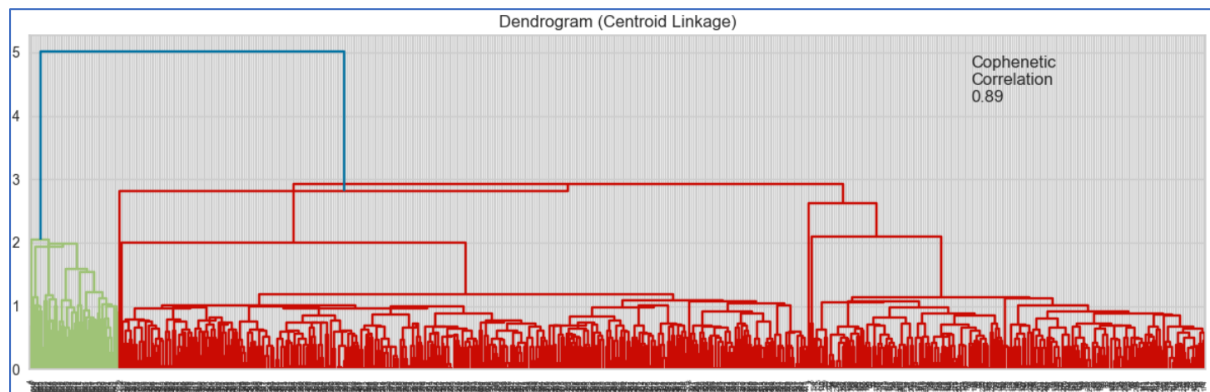


Fig 27. Dendrogram: Centroid Linkage

Dendrogram: Ward Linkage

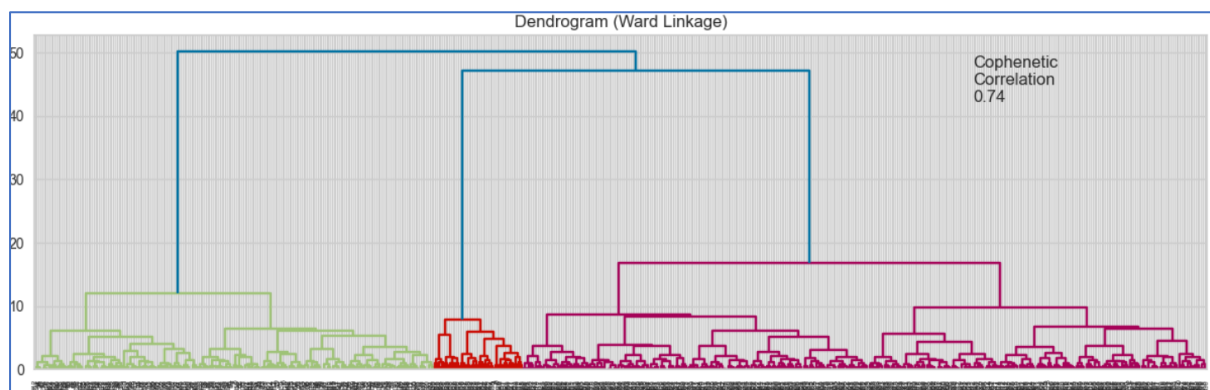


Fig 28. Dendrogram: Ward Linkage

Dendrogram: Weighted Linkage

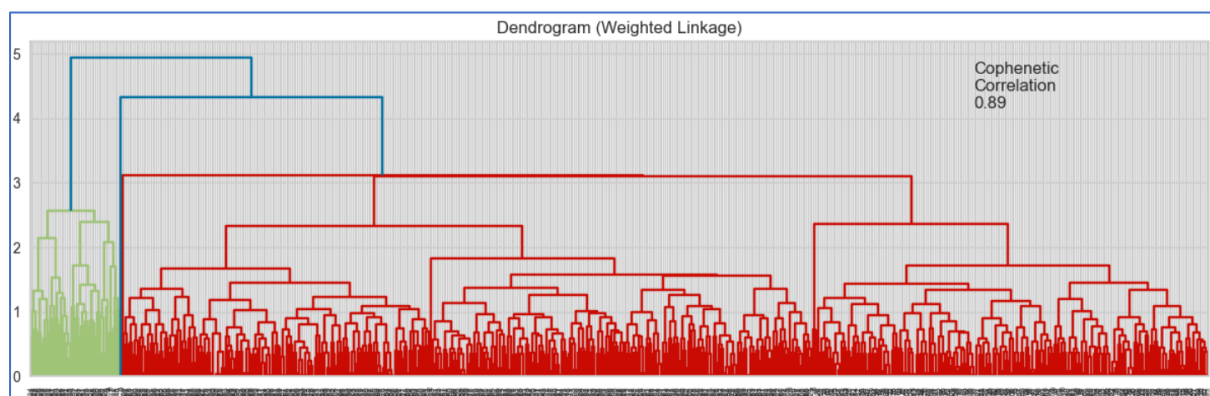


Fig 29. Dendrogram: Weighted Linkage

Cophenetic correlation for different linkage methods

	Linkage	Cophenetic Coefficient
0	single	0.739122
1	complete	0.859973
2	average	0.897708
3	centroid	0.893939
4	ward	0.741516
5	weighted	0.886175

Table 11. Cophenetic correlation for different linkage methods

OBSERVATIONS:

- The cophenetic correlation for the Euclidean distance metric with average linkage was 0.897, while it was 0.74 with Ward linkage. Despite Ward linkage yielding a lower cophenetic correlation, it produces distinct and visible clusters.
- Therefore, we will proceed with the **Euclidean distance** metric and **Ward linkage**.
- From the chosen metrics, we can infer that the number of clusters for obtaining the final model is 4 clusters.

In clustering, outliers can skew results, especially with distance metrics like Euclidean. Using City Block Distance (Manhattan distance) and Mahalanobis Distance can provide a more stable clustering outcome. As there are few outliers in the data set, we will check the dendrograms for Manhattan Distance and Mahalanobis Distance.

Dendrogram: Mahalanobis-Average Linkage

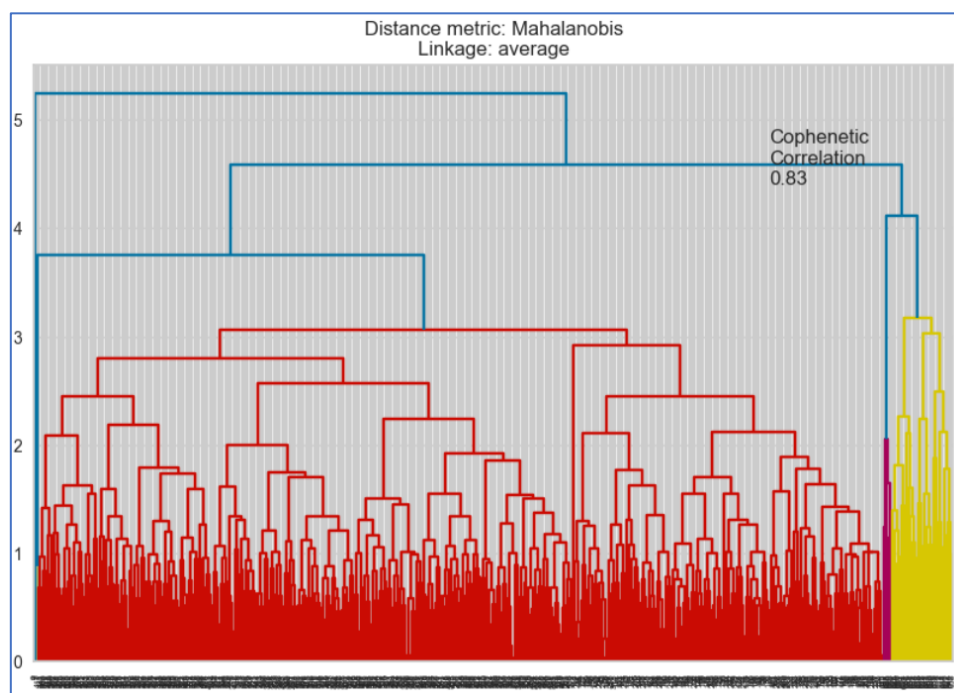


Fig 30. Dendrogram: Mahalanobis-Average Linkage

Dendrogram: Mahalanobis-Weighted Linkage

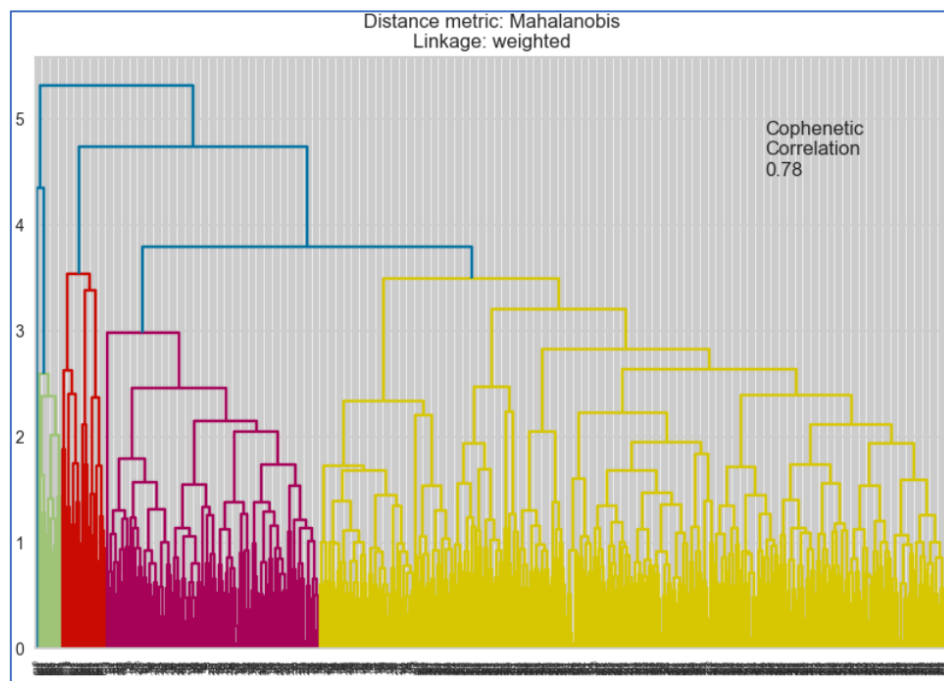


Fig 31. Dendrogram: Mahalanobis-Weighted Linkage

Dendrogram: Cityblock-Average Linkage

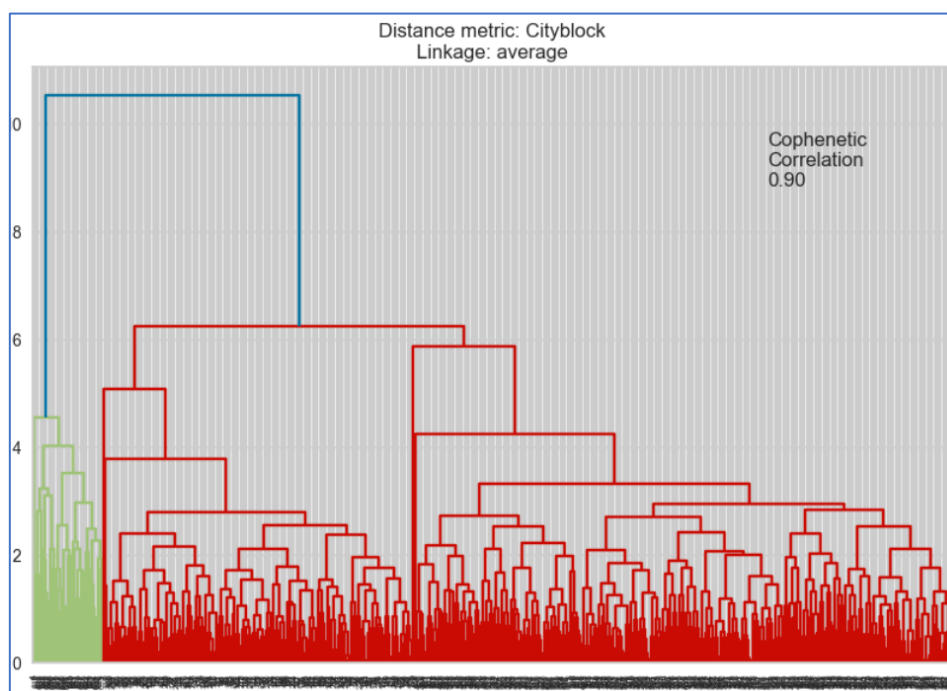


Fig 32. Dendrogram: Cityblock-Average Linkage

Dendrogram: Cityblock-Weighted Linkage

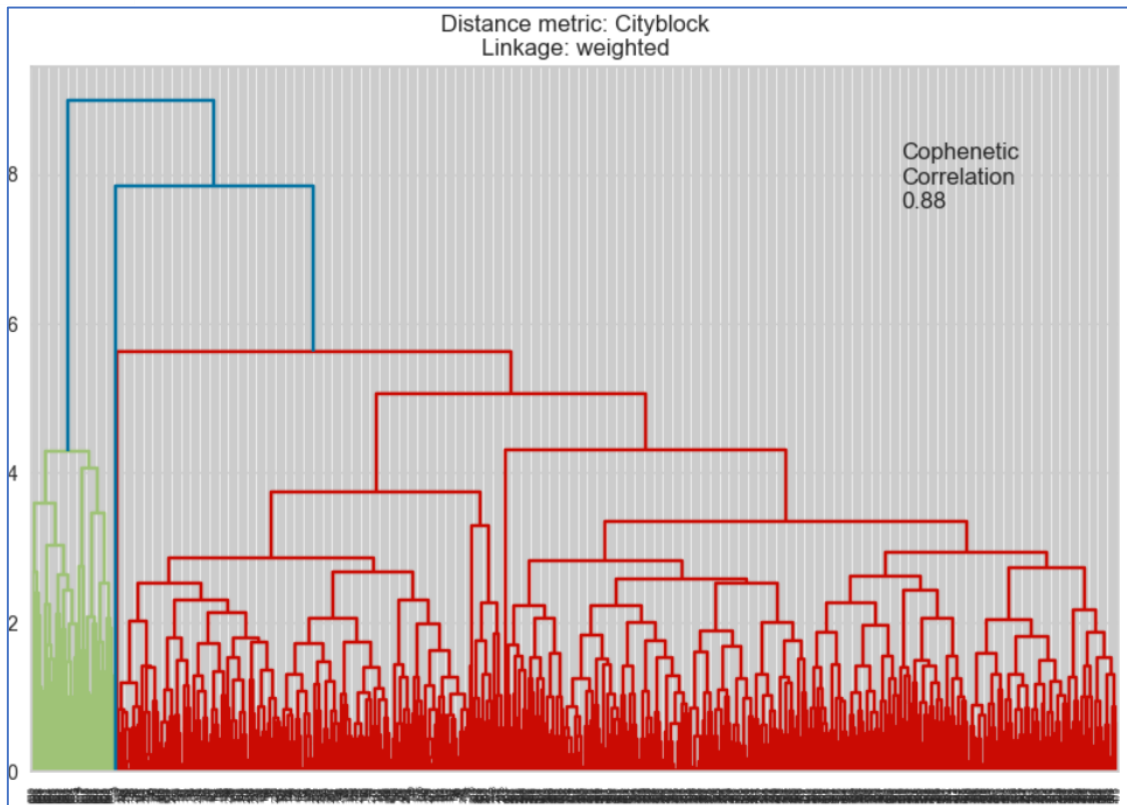


Fig 33. Dendrogram: Cityblock-Weighted Linkage

OBSERVATIONS:

- The dendrograms analysed above show a strong cophenetic correlation value.
- However, the clusters lack distinct separation and appear cluttered.
- Thus, we will proceed with the Euclidean distance metric and Ward linkage.
- Based on the selected metrics, we can determine that the optimal number of clusters for the final model is 4.

6.2.3 Creating Final Model

With **k chosen as 4**, we will assign Agglomerative Clustering cluster labels to the original dataframe and create a final model by using the **AgglomerativeClustering** algorithm from **sklearn.cluster**.

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	HC_Clusters
0	100000	2	1	1	0	1
1	50000	3	0	10	9	0
2	50000	7	1	3	4	1
3	30000	5	1	1	4	1
4	100000	6	0	12	3	2

Table 12. Agglomerative Clustering: Cluster label assigned original dataframe

6.2.4 Cluster profiling

Cluster profiling is done by grouping the data based on the clusters formed and checking the mean, median and boxplot of each variable with respect to the clusters. In order to identify distinct customer segments based on their spending habits, the **Avg_Credit_Limit** variable is used for grouping.

6.2.4.1 Cluster Profile using mean

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segments
HC_Clusters						
0	12151.111111	2.422222	0.937778	3.546667	6.857778	225
1	38298.429319	5.670157	2.523560	0.947644	2.099476	191
2	141040.000000	8.740000	0.600000	10.900000	1.080000	50
3	29474.226804	5.365979	4.448454	1.010309	1.891753	194

Table 13. Agglomerative Clustering: Cluster profiling using mean

6.2.4.2 Cluster Profile using median

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segments
HC_Clusters						
0	12000.000000	2.000000	1.000000	4.000000	7.000000	225
1	37000.000000	6.000000	3.000000	1.000000	2.000000	191
2	145500.000000	9.000000	1.000000	11.000000	1.000000	50
3	23000.000000	5.000000	5.000000	1.000000	2.000000	194

Table 14. Agglomerative Clustering: Cluster profiling using median

6.2.4.3 Cluster Profile using boxplot

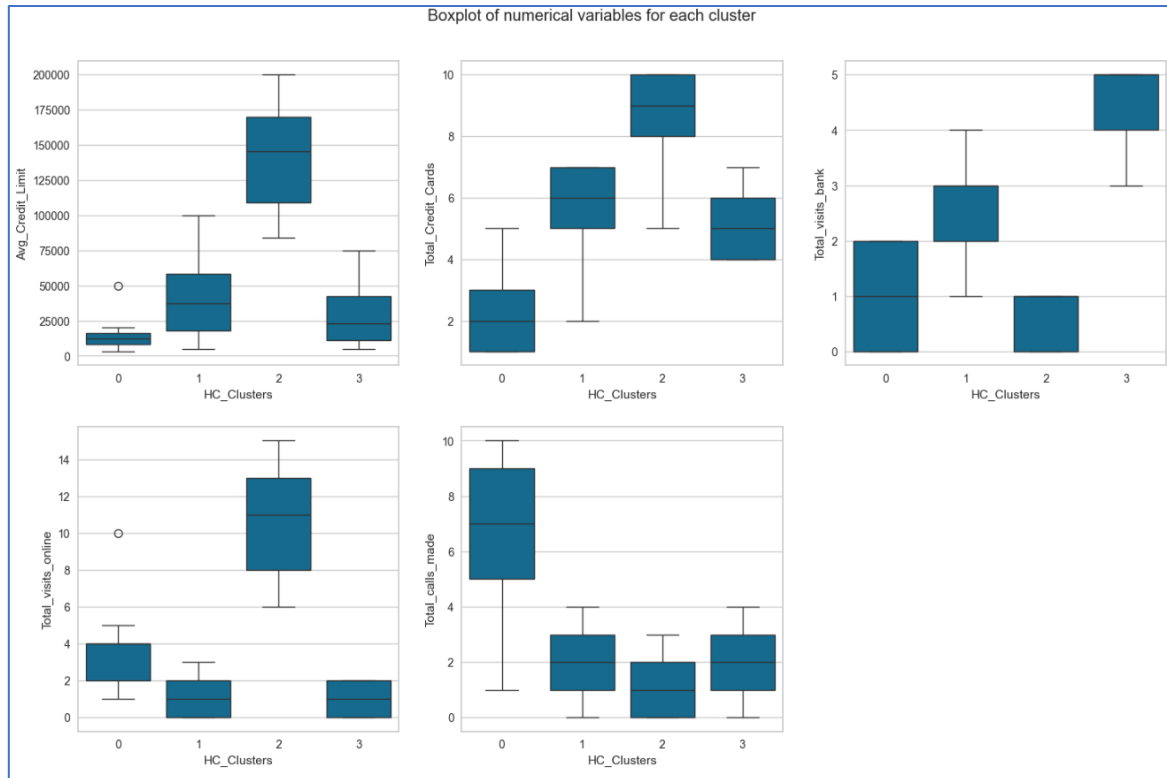


Fig 34. Agglomerative Clustering: Cluster profiling using boxplot

OBSERVATIONS:

Cluster 0

- This cluster comprises approximately 225 customers, accounting for nearly 33% of all credit card holders, the largest percentage among the clusters.
- With an average credit limit of 12,000, the lowest in this group, it suggests that these customers may be less experienced or have lower creditworthiness, indicating a potential need for financial education or support in managing credit.
- They have made only 1 bank visit and 4 online visits, suggesting a preference for self-service options.
- Furthermore, they have the highest number of calls made, which is 7, which may point to a need for assistance or clarification regarding banking services.

Cluster 1

- This cluster includes approximately 191 customers, accounting for nearly 29% of all credit card holders.
- Customers in this group have an average credit limit of around 38,000, indicating that they are likely financially stable and trusted by the bank.
- On average, these customers own 6 credit cards, reflecting a preference for diverse financial products.

- They have visited the bank 3 times, while their online engagement is low, with only 1 online visit. This suggests that these customers may not be fully utilizing their cards.
- Additionally, they make only 2 annual calls to the bank, which may indicate a lack of awareness about available services.

Cluster 2

- This cluster consists of only 50 customers, each with an average credit limit of 141,000, indicating that they are highly eligible and experienced with credit cards, reflecting a strong financial background.
- On average, they hold nearly 9 credit cards, the highest among all clusters, suggesting their proficiency in managing credit and a desire to take advantage of various rewards and benefits.
- They have made only 1 bank visit and 1 call, implying that they do not require assistance.
- However, their online activity is significantly high, with approximately 11 visits, likely for transactions and bill payments.

Cluster 3

- This cluster includes about 194 customers, almost 28% of the customers having credit cards.
- This cluster includes customers with an average credit limit of approximately 29,000. The average credit limit suggests these customers are likely to have a good credit history.
- Customers typically own 6 credit cards on an average. They possibly use multiple cards for different purposes.
- They have visited the bank 5 times, but their online engagement is quite low, with only 1 online visit. This suggests the lack of confidence in online banking or a preference for personal service.
- Additionally, they make just 2 annual calls to the bank. It could suggest satisfaction with current services or a lack of awareness about additional offerings.

6.3 K-means vs Hierarchical Clustering

K-means Cluster Profile

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segments
KM_segments						
0	31226.315789	5.478947	4.515789	1.010526	1.936842	190
1	35904.522613	5.522613	2.477387	0.969849	2.075377	199
2	141040.000000	8.740000	0.600000	10.900000	1.080000	50
3	12167.420814	2.393665	0.927602	3.574661	6.923077	221

Table 15. K-means Clustering Cluster Profile

Agglomerative Hierarchical Clustering Cluster Profile

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	count_in_each_segments
HC_Clusters						
0	12151.111111	2.422222	0.937778	3.546667	6.857778	225
1	38298.429319	5.670157	2.523560	0.947644	2.099476	191
2	141040.000000	8.740000	0.600000	10.900000	1.080000	50
3	29474.226804	5.365979	4.448454	1.010309	1.891753	194

Table 16. Agglomerative Clustering Cluster Profile

OBSERVATIONS:

- **Execution Time:** K-means clustering was typically faster to execute, whereas agglomerative clustering took longer due to the time required to construct dendrograms with different distance metrics and linkage methods.
- **Number of Clusters:** Determining the number of clusters in K-means was challenging, as methods like the elbow method and silhouette scores were not sufficient on their own and the computation time along with the elbow plot and silhouette plot was required. Unlike agglomerative clustering, K-means lack the graphical representation provided by dendrograms.
- **Visualization:** The dendrograms from agglomerative clustering offered excellent visualization of the clusters, aiding in the decision-making process for the number of clusters. The dendrograms gave distinct clusters making it easy to visualize the data into groups.
- **Cluster Evaluation Scores:** The silhouette scores for the clusters were relatively low in comparison (0.374) to the cophenetic correlation, which was approximately 0.76. This makes it challenging to determine the optimal number of clusters in K-means.
- **Clusters:** The clusters exhibit a similar distribution of customers and appear nearly identical. For instance, cluster 0 in K-means corresponds to cluster 3 in agglomerative clustering, and vice versa. The data splits, variable weights, and distributions of the variables are very comparable. Cluster 2 seems to be identical in both K-means and agglomerative clustering.

7. Actionable Insights & Recommendations

- Cluster 2 from both K-means and agglomerative clustering consists of experienced customers with higher credit limits and effective transaction management skills. As these customers frequently utilize online platforms, targeted marketing campaigns can be designed to offer additional benefits related to online transactions, enhancing their engagement with the banking process.
- Cluster 2 is the perfect target group for growing the credit card customer base in the upcoming financial year, as they engage in credit card usage frequently and seamlessly.
- Cluster 0 from K-means and Cluster 3 from agglomerative clustering account for nearly 28% of the population that has made the most bank visits. In today's online banking era, frequent visits may indicate a lack of understanding of online policies and benefits. Therefore, targeted marketing campaigns and educational resources highlighting the advantages of online banking could help expand the customer base and improve understanding of credit card usage.
- The above-mentioned clusters, Cluster 0 from K-means and Cluster 3 from agglomerative clustering represents the target group for organizing online webinars or workshops aimed at educating customers about the advantages of credit card usage and online banking features.
- Cluster 3 from K-means and Cluster 0 from agglomerative clustering together represent nearly 33% of the population, the largest share among the clusters. This group has an average of almost 4 online visits and the highest number of customer calls. By implementing chatbots on the website and mobile app to address common inquiries, we can enhance the overall customer experience and decrease the volume of customer service calls.
- The above-mentioned clusters, Cluster 3 from K-means and Cluster 0 from agglomerative clustering would benefit from the creation of a comprehensive online resource centre that offers FAQs, guides and tutorials on using credit cards and accessing support services. This would help minimize customer service calls, which may be occurring to clarify doubts about transaction processes.
- Cluster 1 from both K-means and agglomerative clustering represents a significant portion of the customer base, with an average of nearly 6 credit cards per customer. However, their bank visits and online interactions are low, averaging only 2. This suggests that these customers may not be fully utilizing their cards. Therefore, it's crucial to implement strategies to boost their engagement.
- The above-mentioned clusters require attractive rewards programs, cashback offers or promotional bonuses for using the card. This can also be achieved by promoting features like mobile payments, contactless transactions and easy online account management.
- It is also important to implement comprehensive training programs for support staff to improve their skills in customer service, problem-solving and product knowledge.
- Establishing a feedback loop where customers can easily report their experiences with support services, enables continuous improvement.
- Regularly reviewing and refining both customer support and marketing strategies ensure they align with customer expectations and market trends.
- Establishing referral incentives that reward existing customers for bringing in new clients can effectively harness the customer base.

By focusing on these solutions, AllLife Bank can improve customer perception of support services, effectively penetrate the market, and enhance the overall customer experience in the coming financial year.